

Geophysics 3A

Potential Fields & Geothermics

Workshop Booklet

2026

If lost, please return to: _____

Workshop Booklet Guide & Assessment

Student Name: _____

Student ID: _____

Purpose of this booklet

This booklet is your working record for workshop activities throughout the course. It is intended to:

- support your understanding through active participation,
- provide a space to develop and record your thinking,
- serve as a study resource for tests and exams.

You are expected to bring this booklet to every workshop and use it actively during group activities.

How workshops are assessed

Workshop performance is assessed using two complementary components:

1. Participation (major component)

Your participation is assessed during workshops based on your engagement with activities and peers. This includes:

- contributing ideas and reasoning during group work,
- engaging constructively with others' ideas,
- asking questions and working through uncertainty.

Participation is recorded throughout the semester. You do not need to speak constantly, but you should be actively involved in your group's work.

2. Booklet (supporting component)

This booklet provides evidence of your engagement with workshop material.

During the semester, your booklet will be collected **during tests** for a quick review. Assessment will focus on:

- completeness of activities,
- evidence of working, reasoning, and annotations,
- overall engagement with the material.

Submission for assessment

- Bring your booklet to each test when requested.
- Booklets will be collected at the start of the test and returned at the end where possible.

If you are unable to submit your booklet at the test:

- You must upload clear photos/scans within 24 hours.
- If no submission is received, the booklet will be marked as incomplete for that checkpoint.

Expectations

This booklet is not marked for perfection or neatness. Instead, it should show:

- your attempt to work through problems,
- annotations, sketches, and intermediate steps,
- corrections and developing ideas.

Working collaboratively is expected; however, your booklet should reflect your own engagement with the material.

Key idea

Workshops are designed to help you think, discuss, and explore ideas. Your grade reflects both:

- how you participate in the learning process, and
- the evidence of that engagement captured in this booklet.

Contents

Workshop Booklet Guide & Assessment	1
Week 1 — Gradients, Curvature, and Potential-Field Intuition	5
Week 2 — Physical Properties, Mixing Laws, and Asking the Right Questions	22
Week 3 — Exploring & Understanding Inversion	36
Week 4 — Gravity: Density, Structures, and Non-Uniqueness	54
Week 5 — Isostasy: Ocean–Continent Structure & Isostatic Gravity	69
Week 6 — Magnetism: Anomalies and Susceptibility	78
Week 7 — Fourier Transforms and Potential Fields	91
Week 8 — Geothermics: Steady-State Heat Flow	96
Week 9 — Geothermics: Transient Heat Flow	107
Week 10 — Geothermics: Finite Differences	118

Student Name: _____

Student ID: _____

Instructions (Instructor Use)

Circle the level that best reflects the student's overall performance since the previous checkpoint. Use the booklet as supporting evidence, but prioritise observed participation.

Criterion	HD	D	C	P	F
Preparation & Contribution	Insightful, advances discussion	Regular, clear reasoning	Occasional, appropriate	Basic, prompted only	Little or none
Engagement with Others	Synthesises ideas, elevates group	Builds on others' ideas	Responds to peers	Listens, limited interaction	Disengaged
Curiosity & Thinking	Explores ambiguity, asks deep questions	Engages with complexity	Asks genuine questions	Attempts answers when prompted	Avoids engagement
Booklet Evidence	Complete, thoughtful, annotated	Mostly complete, good detail	Adequate, some gaps	Minimal effort evident	Sparse or missing

Overall Performance

HD

D

C

P

F

Notes (optional)

Week 1 — Gradients, Curvature, and Potential-Field Intuition

Duration: 3 hours

Setting: Workshop room

Preparation

Pre-reading: Course Notes Ch. 1, 3–4

Lecture videos:

Notes:

Tips

- Work in mixed-background groups; geologist + physicist + ...
- Share your expertise freely and use each other's knowledge to fill gaps in your own.
- Work collectively and constructively towards the common aims. Value each other's opinions, feel free to respectfully discuss differences of ideas.

Purpose

The purpose of this workshop is to build an intuitive and practical understanding of gradients and Laplacians as tools for interpreting geophysical fields. Using familiar examples from topography, you will connect physical intuition (slope, steepness, direction, and curvature) to their mathematical descriptions (partial derivatives, gradients, and the Laplacian), and begin to see how these ideas extend to abstract potential fields used in geophysics.

This workshop also introduces back-of-the-envelope (BOTE) reasoning as a core skill for estimating physical quantities, assessing plausibility, and developing quantitative intuition before applying more formal calculations or computational methods.

Learning Outcomes

By the end of this workshop, you should be able to:

- Describe the gradient of a scalar field in both intuitive (slope and direction) and mathematical (partial derivatives and vector representation) terms.
- Interpret maps and surfaces by identifying regions of high gradient magnitude and explaining their physical significance.
- Conceptually explain curvature (Laplacian) as changes in slope and identify where it is large or near zero in maps or surfaces.
- Construct and interpret gradient vectors graphically from contour maps.
- Apply simple back-of-the-envelope calculations to estimate physical quantities and justify assumptions using order-of-magnitude reasoning.
- Connect the concepts of gradients and curvature to the behaviour of potential fields and their role in geophysical interpretation.

Session Flow (170 min)

0–15 min: Course introduction

What the course covers, weekly rhythm, assessment structure, expectations, office hours, and where to find materials. Download materials while instructor goes over outline of course.

15–60 min: Activity 1.A: Tectonic Plate vs. Freight Train

Work in groups of 3–4 on the back-of-the-envelope style problem.

To estimate these parameters, you will need to know/learn a few things about the structure of Earth, the physical properties of common rock types within each layer, and make assumptions (understand ranges) for required values. You can refer to the *Geology for Geophysicists* notes. You will need to keep track of units and become comfortable with uncertainty and order-of-magnitude estimation.

60–80 min: Mini-lecture and Q & A

Address questions from lecture; otherwise give a concise teaser: scalar vs vector fields, ∇h as direction of steepest increase, and ∇^2 as curvature, vector field = gradient of scalar potential ($F = \nabla\phi$).

80–85 min: Break**85–100 min: Activity 1.B: Sheet Model of Potential Fields**

Hands-on demonstration of a scalar field using a stretched sheet. Boundary heights are fixed (Dirichlet boundary conditions), while pushing up/down on the sheet introduces localized sources and sinks.

Focus on how the gradient (slope) and curvature (Laplacian) respond to boundary conditions and internal forcing, and how increasing observation distance smooths the field.

100–130 min: Activity 1.C: Gradient Activity

Work in groups of 3–4 on the gradient activity.

130–145 min: Activity Debrief

Discuss the answers to the Gradient Activity.

145–155 min: Preview: Physical properties

Next week, we'll examine what I like to call the *geophysical fudge factors*: the physical properties of Earth materials. These properties control how fields are generated, modified, and transmitted. Mathematically, they often appear as scale factors that determine magnitudes and ensure the units make sense—but physically, *properties contain the geology*.

As geophysicists, we measure fields but interpret physical properties, so it is essential to understand how these properties vary with composition and state variables such as pressure and temperature.

We'll also look at how to estimate rock properties using different approaches, including deriving bulk properties from mineralogy. This approach introduces mixing models, and we'll discuss when different mixing relationships are appropriate.

Workshop Notes

Workshop Notes

Activity 1.A

Tectonic Plate vs. Freight Train

1 Purpose

The purpose of this activity is both a warm-up and an introduction to back-of-the-envelope (BOTE) problem solving. So called a BOTE calculation because it is one that can be done relatively simply on a small piece of scrap paper. Such calculations often result in order-of-magnitude estimates, are a way to place rough bounds on a value, and can be used test simple hypotheses before more complex computations are required. It is a useful skill because it can be done in the field to do simple interpretations of data or during a chat with colleagues over drinks/dinner.

2 How to do BOTE calculations

Work backwards. Start with the last equation you will need to solve the problem. Write down what you know and identify what you don't. Often you will need to do multiple calculations on the way to solving the problem.

A couple things to keep in mind:

round quantities – we are generally looking for order-of-magnitude estimates—calculations you can do in your head—not precise answers;

estimate unknowns – few values are truly unknown, bounds even rough can often be set;

use geometric means – you will find that when using bounds values tend towards the smaller bound, a geometric mean $g = (\text{lower} \times \text{upper})^{1/2}$ of the bounds is often closer to the truth than an arithmetic mean;

use scientific notation – quicker for many unit conversions, multiplication and division;

dimensional analysis – always check your units, it can reduce the number of mistakes and can be a way to construct equations when you can't remember the exact formula

2.1 Rounding

We can round numbers to make calculations quicker and easier. For example, how many seconds are there in a year? The exact answer requires the number of seconds in a minute, number of minutes in an hour, number of hours in a day, and number of days in a year (don't forget leap years). That's too complicated. The number is approximately 3.15×10^7 s.

There are two values I keep in my head, 3×10^7 and $\pi \times 10^7$ s. The former is simple and rounded to one digit whereas the latter often useful for canceling π , which shows up in a lot of equations.

2.2 Estimating Unknowns

Even when we have little information about the value for a property or a quantity, we can develop an estimate or place some rough bounds. It is rare that we truly have no idea.

Consider the velocity of a tectonic plate, which you will need for the eventual questions. We know the velocity of a tectonic plate is between 0 and we can easily outwalk it, so we know it is less than m s^{-1} . We can do better. Let's look at 3 ways to estimate it:

Case 1: Start with a random fact you might know.

You might have heard that it is about the rate that fingernails grow. Personally, I can't stand my fingernails to get long. I cut about a millimeter off every two weeks. There are 26 fortnights in a year so

that's a velocity of 26 mm a.

Case 2: Estimate it from something that is easy to determine.

Africa and North America split from Pangea about 180 Ma. Looking at a map, I can estimate about 60° between Morocco and Florida. At the equator (close enough) the distance per degree is ~111.2 km. So there is about 6600 km between the two countries today. That would be an average rate of approximately $(6600 \text{ km}) / (180 \text{ Ma}) \approx 37 \text{ km Ma}$, or an equivalent of 37 mm a.

Case 3: Estimate the bounds if I don't know a reasonable average.

Some will be faster and some slower, so a value between say 10 and 100 mm a⁻¹. Many natural processes result in log-normally distributed values—especially those that vary by orders of magnitude—which means they are skewed towards lower values but have long tails on the high end. In such cases it makes more sense to use a geometric mean to calculate an average than an arithmetic mean. The geometric mean of 10 and 100 is simply the square root of their product, in our plate velocity example,

$$v_g = \sqrt{(10 \text{ mm a}^{-1})(100 \text{ mm a}^{-1})} = \sqrt{(1000 \text{ mm a}^{-1})} \approx 30 \text{ mm a}^{-1},$$

which in this case turns out to be about right.

2.3 Scientific notation

Scientific notation converts every number into a decimal multiplied by a power of 10, i.e., $m \times 10^n$, where m is a decimal value and n is an integer. The number m is called the mantissa or significand and n is called the order or exponent. When multiplying two numbers in scientific notation, we multiply the mantissa and add the orders,

$$(m_1 \times 10^{n_1})(m_2 \times 10^{n_2}) = (m_1 \cdot m_2) \times 10^{n_1+n_2}.$$

Division likewise is easier,

$$\frac{m_1 \times 10^{n_1}}{m_2 \times 10^{n_2}} = \frac{m_1}{m_2} \times 10^{n_1-n_2}.$$

Because we often only care about order-of-magnitude rather than precise estimates, we can limit our m_1 and m_2 to one to two digits.

Many unit conversion become a sinch in scientific notation. Using our seconds in a year above, if we needed the number of seconds in a million years (as geoscientists we often work in deep time) we multiply by 10⁶. Hence, $(3 \times 10^7 \text{ s a}^{-1})(10^6 \text{ a Ma}^{-1}) = 3 \times 10^{13} \text{ s Ma}^{-1}$.

2.4 Dimensional analysis as a test

Dimensional analysis is the process of checking units during mathematical operations to ensure that the correct units are determined in the end. For example, if you calculate the force of gravity from $F = mg$, how do you know that you end up with the appropriate unit of newtons (N)? We can write this as an equation:

$$\begin{aligned} F &= mg \\ [\text{N}] &\stackrel{?}{=} [\text{kg}][\text{m s}^{-2}] \\ [\text{N}] &= [\text{kg m s}^{-2}] \end{aligned}$$

Of course, this can only be verified as correct if you know that a newton is defined as a kg m s^{-2} , so it is useful to know how units are defined in terms of base units like m, s, kg, C, etc.

Let's look at another example, the calculation of pressure. Such a calculation is commonplace for geoscientist. Often we report pressures in mega-pascals, MPa. The equation for pressure is simple,

$$P = \rho gh,$$

where ρ is the average density of rock above the point of interest, g is the gravitational acceleration, and h , is

the thickness of rock above. Density is typically reported in kg m^{-3} , gravity in m s^{-2} , and height in km. Note neither pressure or height are reported in standard SI units. So let's see if this works dimensionally,

$$\begin{aligned} [\text{MPa}] &\stackrel{?}{=} [\text{kg m}^{-3}][\text{m s}^{-2}][\text{km}] \\ [\text{MPa}][\text{Pa MPa}^{-1} \cdot 10^{-6}] &\stackrel{?}{=} [\text{kg m}^{-2} \text{s}^{-2}][\text{km}][\text{m km}^{-1} \cdot 10^{-3}] \\ 10^{-6}[\text{Pa}] &\stackrel{?}{=} 10^{-3}[\text{kg m}^{-1} \text{s}^{-2}], \end{aligned}$$

since a Pa is a N m^{-2} or a $\text{kg s}^{-2} \text{m}^{-1}$, we can see

$$10^{-6}[\text{Pa}] \neq 10^{-3}[\text{Pa}].$$

So, we are missing a conversion factor of 10^{-3} to give our result with units in MPa.

Question 1. Tectonic Plate versus Freight Train

For this problem, I want you to answer two questions:

- (a) Which has larger momentum? Momentum is given by $p = mv$, where m is mass and v is velocity.
- (b) Which has larger kinetic energy? Kinetic energy is given by $K = \frac{1}{2}mv^2$.

Remember to work backwards. Start with mass, how do you estimate it for a tectonic plate, a freight train? Google won't be much help because it cannot tell you the answer (and often gives one orders of magnitude off when students ask). You'll need to do some out of the box thinking, recall some things you might know and some things you might not, for instance, what comprises a plate, not just horizontally, but vertically?

You can use Google for some basic geology questions, but don't ask it for numbers/quantities.

Use this page to complete your BOTE estimates.

Activity 1.B

Sheet Model of Potential Fields

3 Purpose

This short activity uses a physical sheet model to develop intuition for scalar fields, gradients, and curvature. By fixing boundary values and introducing localized perturbations, you will explore how fields respond to boundary conditions and internal sources, and how these effects relate to geophysical potential fields.

4 Setup

Work as a group around a stretched sheet. Some students will hold the edges, while others gently push or pull on the sheet from below or above.

5 Tasks and Prompts

Question 1. Boundary conditions (Dirichlet)

Hold the edges of the sheet at fixed but varying heights and pull it taut.

- (a) How do the boundary values control the shape of the interior of the sheet?

- (b) Are there any local maxima or minima within the interior?

Question 2. Sources and sinks (Poisson region)

Introduce localized perturbations by pushing up (source) or pulling down (sink) at points within the sheet.

- (a) How does the interior of the sheet respond to these perturbations?

- (b) Do these changes affect the boundary? Why or why not?

Question 3. Gradients and curvature

Consider the geometry of the sheet as a surface.

- (a) Where is $|\nabla\Phi|$ largest? How can you tell from the shape of the sheet?

- (b) Where is $\nabla^2\Phi \neq 0$ versus ≈ 0 ?

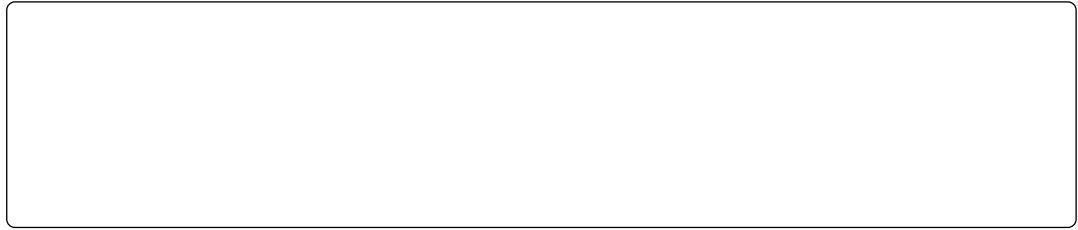
- (c) Which features correspond to regions of strong curvature?

Question 4. Effect of observation distance

Imagine observing the sheet from farther above.

- (a) Why does the surface appear smoother as the observation distance increases?

- (b) Which features disappear first, and why?



6 Key idea

Gradients describe how quickly and in what direction a field changes, while curvature describes how those changes themselves vary in space. In geophysical potential fields, boundary conditions and internal sources control the field, and increasing observation distance smooths short-wavelength features.

Activity 1.C

Gradients Activity

"I will never listen to ocean waves or view a beautiful sunset in quite the same way again. That is perhaps the greatest gift one can gain by delving into calculus: It is a whole new way of looking at the world, accessible only through the realm of mathematics."

— Jennifer Ouellette, *The Calculus Diaries: How Math Can Help You Lose Weight, Win in Vegas, and Survive a Zombie Apocalypse*

7 Introduction

As we study geophysical theory it will become clear that in order to unlock the secrets of the physical world, we must understand the changing nature of fields. If the fields were constant the world would be very boring indeed, for it would not contain any variations in physical properties that we could remotely sense. Variations in physical fields help us better understand the state (temperature, pressure, and composition), structure, and/or the evolution of our wondrous planet. One of the simplest ways to explore a field is to examine the rate of change of the field, or gradient, which is really a fancy way of saying slope.

In this workshop you will explore conceptually the physical meaning of a gradient using intuitive examples from elevation. We will be able to see how the gradient depends upon the direction we consider and how a derivative can be calculated simply for a complex geometry, and how the gradient of a scalar field such as elevation results in a vector field requiring both magnitude and direction in order to describe the gradient.

We will start the exercise by building a conceptual understanding of the gradient before creating a more complete mathematical description.

8 Mount Shasta Example

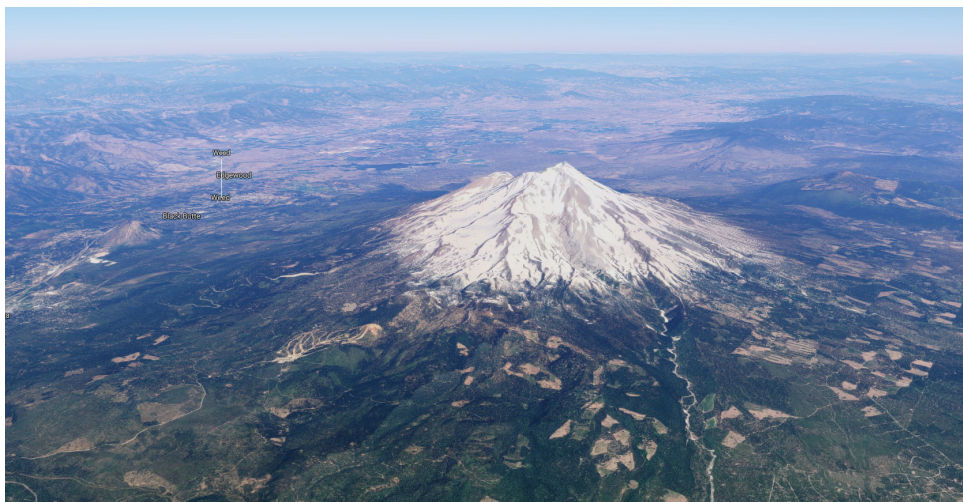


Figure 1: Mount Shasta, California (image from Google Earth).

Elevation is a good place to start to think about gradients because we all intuitively know what a slope means in this context. As far as topography goes, Mount Shasta is pretty simple. It is a stratovolcano at the southernmost end of the associated with the Cascade Volcanic Arc, and looks like a textbook example of a volcano in that it is fairly symmetric radially and sits in the middle of a vast flat plain.

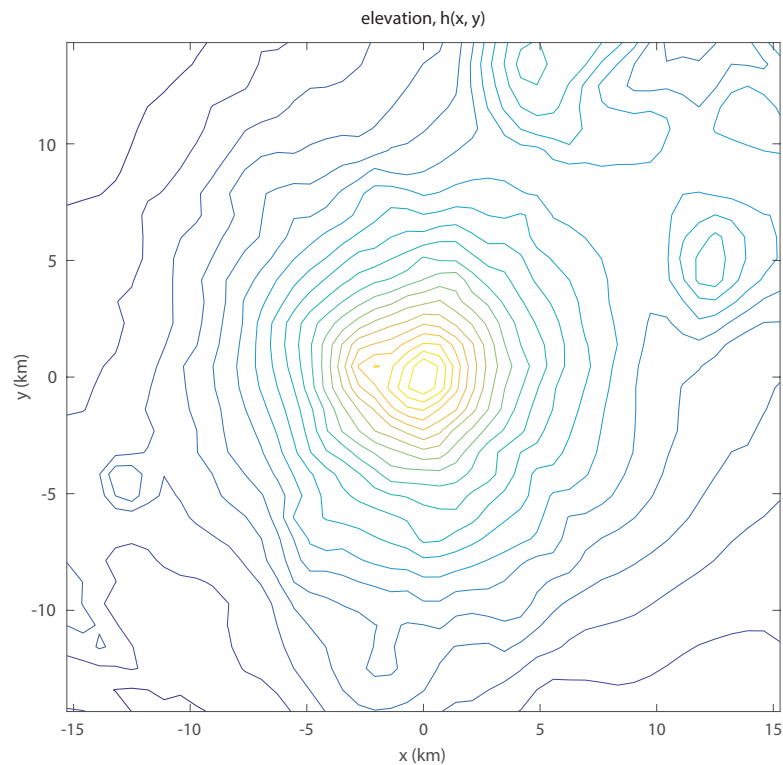
Question 1. Shasta gradient

Figure 2: Topographic map of Mount Shasta.

- (a) Over the topographic contours on Figure 2, draw a series of arrows pointing up slope representing the opposite direction water might flow. Make the arrows longer when the slope is steeper and make the arrows shorter when the slope is gentler.

- (b) How did you decide to draw the direction of the arrows?

- (c) How did you decide to draw the length of the arrows?

- (d) Why does it require two quantities, length and direction, to describe the gradient?

9 The mathematics of gradients: the slopes of Mt. Shasta

Consider hiking a profile line over the peak of Mt. Shasta (Figure 3, top panel). You will start by walking up the slope of Mt. Shasta. The slope starts gently going up in elevation (a positive slope in Figure 3, bottom panel), which increases as you climb in elevation. The slope increases and reaches a maximum near the peak, then the slope starts to decrease as elevation still increases. Once you reach the snow-capped peak there is no change in slope (i.e., slope = 0). Then you decide to pull out your sled and quickly descend the other side (negative slope).

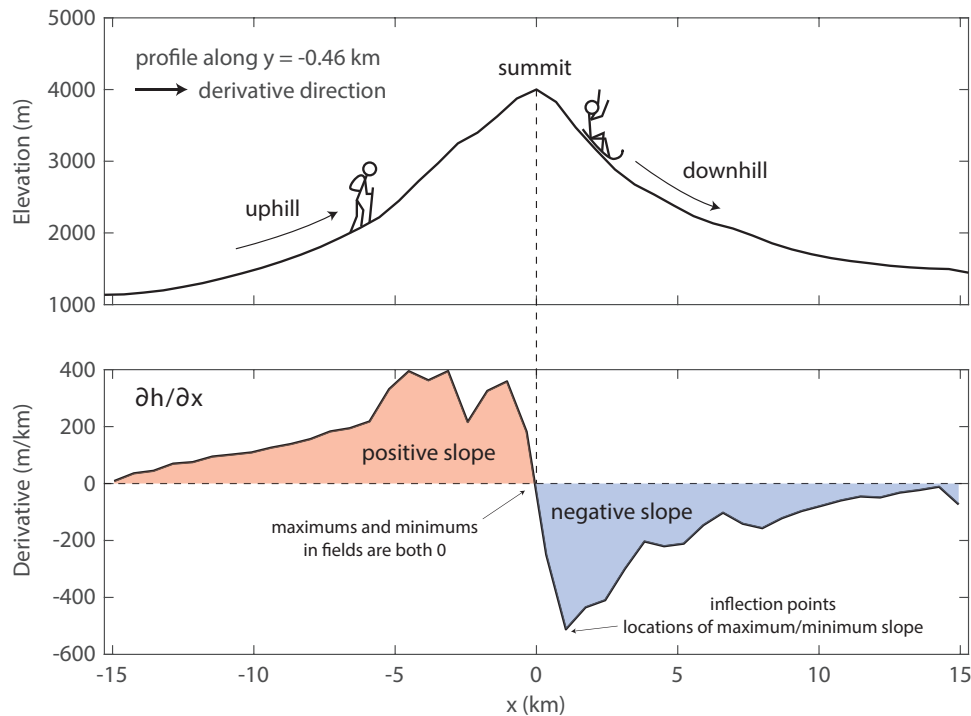


Figure 3: Elevation profile (top) and gradient along the elevation profile (bottom).

What we want to do is be able to describe the changes in slope that you as a hiker feels mathematically. Say every step you take is a similar distance laterally (the run) and the height you go up (or down, negative up) in elevation (the rise) allows us to express the slope as

$$\text{slope} = \frac{\text{rise}}{\text{run}}. \quad (1)$$

But the direction that you take matters. Since we don't necessarily know the direction of the maximum slope we'll force the hiker to walk in just one direction, say west to east (+x) to start. The size of the steps they take are Δx . In this case, we can express the slope as

$$\frac{\partial h}{\partial x} \approx s_x = \frac{h(x + \Delta x) - h(x)}{(x + \Delta x) - x} = \frac{h(x + \Delta x) - h(x)}{\Delta x}. \quad (2)$$

At this point the way we solve this mathematically as opposed to a computer are different. A computer will solve the problem much like above, taking the difference between the heights at $h(x + \Delta x)$ and $h(x)$ and dividing by the distance of each step (Δx).

For our purposes generally during the course, we will take a more mathematical approach, taking the case where we want to know the slope at a continuous series of point along the elevation profile. In this case the distances between our steps approach zero and the slope we approximate the approaches a limit of the true slope at each point. This is the definition of the derivative, and you can find an extended discussion as a tutorial online and in the math appendices in the course notes.

If instead, you were to hike from south to north (+y-direction) we can write the slope as

$$\frac{\partial h}{\partial y} \approx s_y = \frac{h(y + \Delta y) - h(y)}{(y + \Delta y) - y} = \frac{h(y + \Delta y) - h(y)}{\Delta y}. \quad (3)$$

If you were to walk in a series of profiles in x or y, you could map out the slope with respect to that series of traverses. You will notice that the slope you get is very different depending upon whether you walked in the x-direction or the y-direction ¹. We can write the function of the slope, s at each location (x, y) as

$$\mathbf{s}(x, y) = s_x \hat{\mathbf{x}} + s_y \hat{\mathbf{y}}, \quad (4)$$

where the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are a unit vectors in the x- and y-directions, respectively. Note that the slope is a vector quantity, which means it has both a magnitude and a direction. In calculus, we describe the gradient (∇) of the elevation function, $h(x, y)$, by

$$\nabla h = \frac{\partial h}{\partial x} \hat{\mathbf{x}} + \frac{\partial h}{\partial y} \hat{\mathbf{y}}. \quad (5)$$

Because the components are in different directions, we cannot simply add them to find the maximum slope.

From the result above, we learn that the slope is different depending upon the direction you walk (N-S) versus (E-W) which confounds the direction of the maximum slope. You may also note that if we had walked in the opposite (negative) directions, we would have changed the sign of our slopes. Do not fear, there is a simple mathematical solution to determine both the maximum slope and its direction.

If we assume that from each point that we walked in either the +x or +y direction we can describe the local surface as a plane, the directions we walked were purposely perpendicular to each other. It turns out that these two directions can be used to define that plane and both the angle and magnitude of the maximum slope (magnitude) along it (Figure ??). To determine the maximum slope along each individual little plane, we use the Pythagorean theorem, which you may remember from high school,

$$\|s\| = (s_x^2 + s_y^2)^{1/2}. \quad (6)$$

Note that because we square each slope component before adding them, we have ensured that our resultant magnitude of the slope is positive. To determine the direction of the maximum slope, we can use trigonometry to determine the angle with respect to the x-axis as

$$\tan \theta = \frac{s_y}{s_x}. \quad (7)$$

Note that to find the appropriate angle we will have to keep track of the signs of s_x and s_y to ensure that θ lies in the appropriate quadrant.

10 Marble Canyon Example

Let's look at another topographic example at Marble Canyon in Arizona. Marble Canyon sits before the entrance of the Grand Canyon. The area surrounding the canyon has relatively flat topography except for some steep-sided cliffs, one east of the canyon that runs approximately north to south and a second, north of the canyon which varies in azimuth. The canyon itself runs northeast to southwest through the center of the image, carved by the Colorado River.

In this exercise, we will take the previous example one step further by determining the slope along each individual axis and then combine them as we might in the mathematical sense and then combine the results to obtain determine a magnitude and direction. Again, we'll do this graphically rather than mathematically to

¹Since Mt. Shasta is fairly radially symmetric, you will notice that the results are pretty similar if you rotate the one of the plots 90° with respect to the other. This is a relatively special case as we can't generally do this, however, many of the problems we will solve in this course either change in one direction only or are symmetric in some way to make the problems simpler.



Figure 4: Marble Canyon, Arizona (image from Google Earth).

gain further insight into how the mathematics works. Note there is a substantial region where the topography is relatively flat (i.e., the slope is negligible). You don't need to put any arrows in these positions so focus on regions where elevation changes.

Question 2. Marble Canyon

- (a) In Figure 5 are two of the same topographic map of Marble Canyon. On the top map, make a series of arrows parallel to the x-axis with length representing the slope in the x-direction. Again, arrow should point up slope. Now on the same topographic map, make a series of arrows parallel to the y-axis with length representing the slope in the y-direction. Once you have both sets of arrows, create a new set of arrows on the bottom topographic map that represent a combination of the two. The resultant vector is simply a graphical application of the Pythagorean equation. What you will have done is the same thing that the math is doing. So when we are working with abstract fields that cannot be seen or felt by us, remember, the concept is shown here is the same.
- (b) How do the individual derivative components behave with respect to the ridge running north east of the canyon?

- (c) For the northern ridge, why does the y-component have only positive slopes whereas the x-component is positive in some places and negative in others?

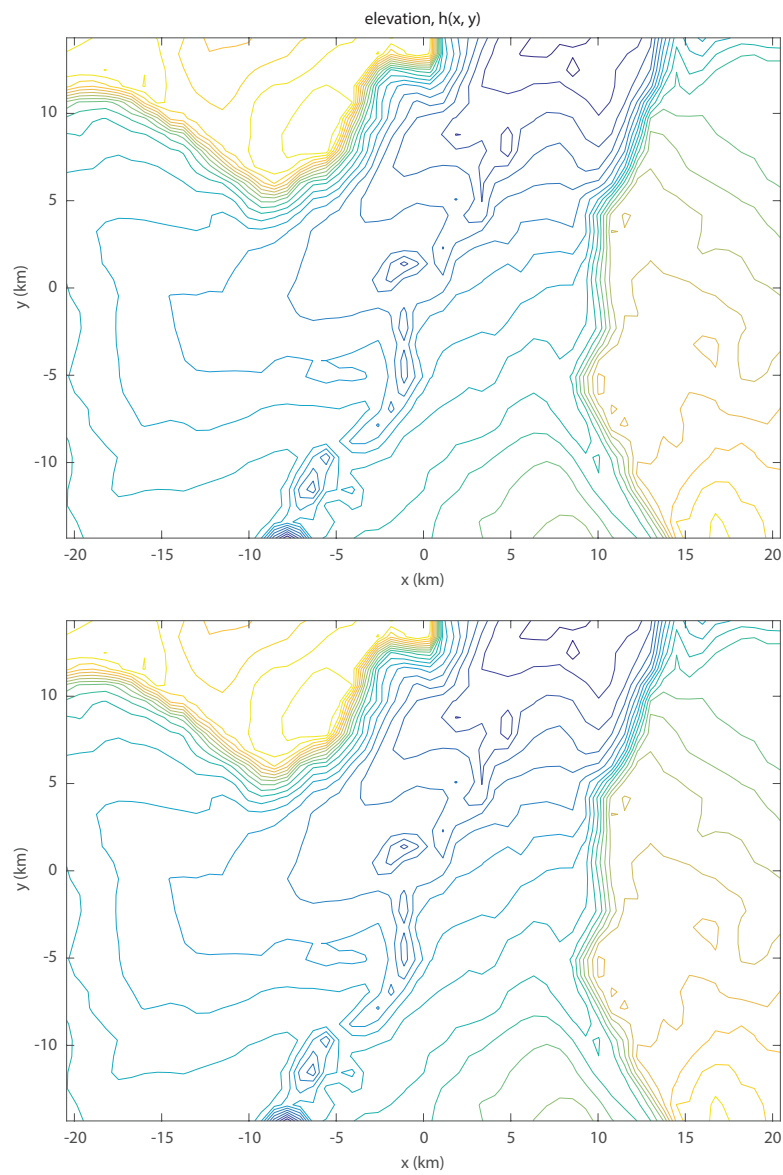


Figure 5: Topographic map of Marble Canyon.

(d) Can you find examples of either of the above situations occurring in the canyon itself?

Week 2 — Physical Properties, Mixing Laws, and Asking the Right Questions

Duration: 3 hours

Setting: Workshop room

Preparation

Pre-reading: Course Notes Ch. 2

Lecture videos:

Notes:

Purpose

This workshop develops intuition for how physical properties (density ρ , thermal conductivity k , thermal diffusivity κ , bulk modulus K_T , shear modulus μ , seismic velocities V_P , V_S , magnetic susceptibility χ , and heat production A) depend on state variables (pressure P , temperature T , composition, porosity/cracks), and how geometry and fabric determine appropriate mixing relations (arithmetic, geometric, harmonic; Voigt–Reuss–Hill). Students practice cross-disciplinary collaboration and formulate the key questions a practicing geophysicist must ask a geologist before modeling or surveying.

Learning Outcomes

- Select and justify mixing laws based on rock fabric (random granular, layered/foliated, fractured) and target property (bulk vs. transport vs. elastic).
- Reason about how P – T –composition/porosity shift properties and create anisotropy.
- Identify which properties most strongly influence gravity, magnetic, and conductive heat-flow interpretations.
- Elicit geological information via well-posed questions to constrain non-uniqueness.

Session Flow (170 min)

Create mixed groups (4–5 students each) with both geology and physics/engineering backgrounds.

0–10 min: What does geophysics see?

Much like our senses, different fields are sensitive to different things. Are geophysical fields sensitive to individual minerals or bulk rock properties? Are they sensitive to different scales?

10–30 min: Mini-lecture and Q&A

- Bulk-additive vs. pathway-limited vs. elastic-interaction properties.
- Mixing relations: arithmetic (parallel), harmonic (series), geometric (random granular), Voigt–Reuss–Hill (elastic bounds).
- Scale-dependent anisotropy (micro/meso/macro fabrics).

30–45 min: Live Demonstration — “Crowd Conductivity”

Aim: Intuitively illustrate why different mixing laws arise and how anisotropy appears.

1. **Random scattering** (students distributed irregularly): volunteer transits across room. Multiple semi-obstructed pathways $\Rightarrow$ geometric-like effective behavior.

2. **Parallel corridors** (lanes aligned with motion): fastest transit $\Rightarrow$ arithmetic mean (parallel pathways).
3. **Series barriers** (rows perpendicular to motion): slowest transit $\Rightarrow$ harmonic mean (series resistances).
4. **Anisotropy**: compare transit parallel vs. perpendicular to fabric.

Optional: time a handful of transits and note distribution shape differences.

45–65 min: Simulation Walkthrough

Builds on crowd conductivity with computation. We'll examine a pre-built Python visualization of transit times through three media (random, parallel, series) and the resulting histogram shapes. Relate to effective properties and mixing choices.

65–95 min: Activity 2.A: Photo-Based Anisotropy (*micro–meso–macro classification*)

Distribute a set of photos (thin section, hand sample/outcrop close-up, outcrop/aerial). For each image, groups identify: geometry (random granular/layered/foliated/fractured), appropriate mixing law for density, conductivity, and moduli, whether the material is isotropic at that scale, and how conclusions change with scale.

95–155 min: Activity 2.B: Scenario Rotations (rich, real-world cases)

Each group tackles two scenarios (20 min each: 12 min discussion, 5 min notes, 3 min question drafting). Use the dedicated scenario handouts. Outcomes must include: (i) property/mixing-law decisions and (ii) 2–3 *questions to ask the geologist* before committing to a model/survey.

155–165 min: Scenario Debrief

Discussion of any new or unresolved questions.

165–175 min: Preview: Inversion

In geophysics, we typically measure fields and infer the underlying physical properties. In this course, however, we mostly focus on the *forward problem*: computing fields from a known distribution of properties.

In practice, we usually face the opposite challenge: the *inverse problem*. Inverse problems are difficult for several reasons:

- they are often *underdetermined*, with fewer observations than unknowns;
- *non-unique*, meaning different models can produce the same data; and
- *unstable*, where small changes in the model can produce large changes in the predicted field.

In addition, many geophysical relationships are non-linear, which adds further complexity.

Next week, we'll begin looking at how inversion methods address these challenges.

Workshop Notes

Workshop Notes

Activity 2.A

Physical Properties & Mixing Laws

Learning Objectives:

Learn how:

- geometry of materials affects bulk physical properties,
- how geometry necessitates the use of different mixing models,
- how scale affects what we perceive, and
- how to visually estimate the behavior of rock properties.

Background

Mixing relationships are mathematical methods that we use to approximate the physical behavior of an rock from its constituent mineralogy and porosity. These relationships can sometimes be derived from first principals—as we'll show for harmonic means and thermal conductivity later in the course—though more often they are empirical approximations that capture the messiness of the world.

Physical properties can be divided into two broad classes.

Bulk properties depend primarily on the amount of each constituent material present and are relatively insensitive to how those materials are arranged. Examples include density (ρ) and specific heat capacity (C_p). These properties are generally estimated using volume-weighted arithmetic averages.

Geometry-controlled properties depend not only on the constituent materials but also on how they are distributed and connected. Examples include thermal conductivity (k), electrical conductivity (σ), and many elastic properties (bulk and shear moduli). For these properties, the appropriate mixing relationship depends on the geometry and orientation of the materials.

Table 1: Guidelines for selecting mixing relationships.

Geometry / Fabric	Density	Conductivity	Elastic Moduli
Random granular mixture	Arithmetic	Geometric	Voigt–Reuss–Hill
Layered, parallel to transport	Arithmetic	Arithmetic	Direction-dependent
Layered, perpendicular to transport	Arithmetic	Harmonic	Direction-dependent
Strongly fractured or porous material	Arithmetic	Depends on connectivity; often harmonic-like	Depends on fracture geometry and filling

Instructions:

Open the slide deck (`isotropy_anisotropy.ppt`) on MyLearning course website. The pictures of rocks range from micrographs to aerial photos. For each photo, ask:

If this were a gravity survey, would the distinction between individual minerals matter?

If this were a conductivity problem, would connectivity matter?

Consider how your interpretation might **change at a different scale**.

For each photo fill in your responses that:

- Identify the **fabric/geometry**.
- Identify the **approximate scale** of the photo.
- Select an appropriate **mixing law for density** (ρ).
- Select an appropriate **mixing law for thermal conductivity** (k).
- State whether the material is **isotropic or anisotropic** at this scale.

Photo	Description	Scale Micro / Macro / Meso	Mixing Law (A)rithmetic, (G)eometric, (H)armonic		(I)sotropic / (A)nisotropic
			Density	Thermal Conductivity	
Example	uniform porous sandstone	Meso	A	G	I
#1					
#2					
#3					
#4					
#5					
#6					
#7					
#8					
#9					
#10					
#11					
#12					
#13					
#14					
#15					

Key Ideas

For bulk rock properties, an arithmetic mean is always used to compute the effective property estimate.

For pathway sensitive properties,

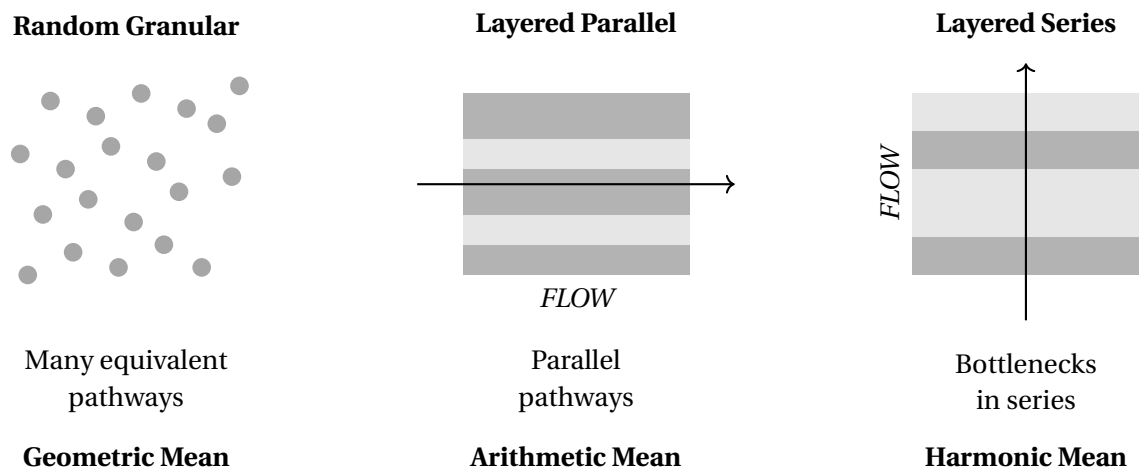


Figure 6: Conceptual relationship between fabric, transport geometry, and appropriate mixing relationship.

Activity 2.B

Modeling Property Scenarios: Choosing Physical Property Estimation Methods

Intructions

Below you will be provided with a set of scenarios that you as a geophysicist are required to develop a physical property model for. The point of this exercise is to develop a model that can be used as a reference based on the available information. In some cases the best and only option may be to run a geophysical survey and model the property from the results, though generally we can use existing samples and datasets to produce an initial property model.

The responses to these scenarios closely align with the lecture notes on methods used to determine/estimate physical properties.

For the discussions, students should be in groups mixed with geologists and physicists. For each scenario:

- draft 2–3 questions that a practicing geophysicist should ask the geologist for additional information about the geologic setting before devising your scenario,
- state your *preferred method* to obtain the required property (e.g., direct measurement, drill-core lab data, composition-based estimate, proxy/correlation, thermodynamic modeling, geophysical inversion/velocity–density regressions), and
- justify why you chose it over other options, list pros and cons, and write 2–3 key *questions to ask the geologist*.

Tip: articulate data availability, scale, P – T effects, alteration/weathering, and non-uniqueness when comparing methods.

Scenario A: Basin Density Model (Resources/Tectonics)

Motivation/Goal. You are supporting a basin analysis that informs *water resource assessment* and subsidence related to extraction. Drill cores are available at multiple wells, and a seismic stratigraphic framework defines a layered structure across the basin. The task is to build a depth-dependent **density model** suitable for forward gravity modeling and basin compaction studies. How will you use the available samples and geophysical data to develop the density model?

Your plan**2–3 Questions for the Geologist****Preferred method and why****Pros**

- _____
- _____

Cons

- _____
- _____

Scenario B: Mountain-Belt Cross-Section (Tectonics)

Motivation/Goal. You are preparing a balanced structural cross-section across a metamorphic belt to support a tectonic reconstruction a for a large civil engineering project (construction of a hydroelectric dam). Rock types are mapped at the surface; access to samples are sparse and uneven, though the region has been explored for mineral resources previously. You need **density** for major units to test isostatic plausibility and gravity fit. How do you produce a density model?

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros

- _____
- _____

Cons

- _____
- _____

Scenario C: Upper-Mantle Density for Isostasy (Geodynamics)

Motivation/Goal. You are part of a team trying to better understand subduction dynamics. A 3-D regional tomography experiment images V_s and V_p within the mantle down to a depth of 410 km, capturing the mantle wedge, subducted plate and remaining upper mantle. The geodynamic model requires an upper-mantle **density** model as an input. What would be your approach to producing this density model?

Your plan

2–3 Questions for the Geologist

Preferred method and why

Pros

- _____
- _____

Cons

- _____
- _____

Scenario D: Aeromagnetic Susceptibility Model (Mineral/Exploration)

Motivation/Goal. For a greenfields **mineral exploration** program, you have a regional aeromagnetic survey. Your brief is to produce a **susceptibility model** that distinguishes key prospective units and to flag where remanent magnetization may invalidate induced-only assumptions.

Your plan**2–3 Questions for the Geologist****Preferred method and why****Pros**

- _____
- _____

Cons

- _____
- _____

Scenario E: Heat-Flow Study in Granitic Terrane (Geothermal/Environmental)

Motivation/Goal. An energy agency is scoping geothermal potential and subsurface thermal risk for infrastructure. You must estimate **thermal conductivity**, $k(T)$ and **heat production** for granitic and metasedimentary units to forecast geotherms and transient responses. A number of pilot drill holes exist in the basin above, but do not intersect the granite. The only information about the granite comes from a gravity survey. How do you estimate the physical properties within the basin and granitic basement?

Your plan**2–3 Questions for the Geologist****Preferred method and why****Pros**

- _____
- _____

Cons

- _____
- _____

Scenario F: Geothermal Heat Flow at the Base of the Antarctic Ice Sheet (Geothermal/Glaciology)

Motivation/Goal. You are tasked with developing a geothermal heat flow model into the base of an ice sheet to support assessment of the ice sheet stability due to recent climate change. Several coastal nunataks provide limited outcrops with physical specimens and geochemistry, but their small areal extent and clustered distribution offer only modest constraints on subsurface lithology and properties beneath the ice interior. There is a relatively low-resolution seismic volume for the region and high-resolution gravity data. How would you develop the necessary **thermal conductivity** and **heat production** models?

Your plan**2–3 Questions for the Geologist****Preferred method and why****Pros**

- _____
- _____

Cons

- _____
- _____

Week 3 — Exploring & Understanding Inversion

Duration: 3 hours

Setting: Lecture room with laptops (sphere inversion app)

Preparation

Pre-reading: Course Notes Ch. 13

Lecture videos:

Notes:

Purpose

This workshop introduces the structure of inverse problems in potential-field geophysics, contrasting forward prediction with inverse estimation. Students explore the role of misfit, data weighting, and regularization, and examine how non-linearity and non-uniqueness arise in gravity modelling. Through a sequence of activities, they classify linear vs. non-linear models, reason qualitatively about non-uniqueness, and engage in hands-on inversion of a buried sphere using grid search and Gauss–Newton methods.

Learning Outcomes

- Distinguish forward and inverse problems and articulate the role of misfit, weighting, and regularization.
- Classify simple geophysical relations as linear or non-linear and propose appropriate linearizations.
- Recognize manifestations of non-uniqueness in gravity and magnetic profiles and propose discriminating data.
- Derive or confirm analytical Jacobian expressions for the buried sphere model.
- Interpret misfit surfaces, inversion paths, and the effect of noise and damping.
- Explain parameter trade-offs and resolution limitations inherent to the buried-sphere problem.

Session Flow (170 min)

Create mixed groups (4–5 students).

0–10 min: Warm-up

What can you learn from a shadow?

- What can you determine directly from the shadow?
- What can you estimate only if additional assumptions are made?
- What information cannot be determined uniquely?
- What prior knowledge are you using when deciding what the object might be?

How does this relate to inversion?

- A shadow does not uniquely determine the object that created it.
- Likewise, a gravity anomaly does not uniquely determine the density distribution that produced it.
- The challenge of inversion is deciding which of many acceptable models is most reasonable.

10–40 min: Mini-lecture — Forward vs Inverse, Misfit, Weighting, Challenges

- Forward vs. inverse models: $m \mapsto g(m)$ vs. $d \mapsto m$.
- Misfit: residual $r = d - g(m)$; weighted least squares $\Phi_d = \|W_d r\|_2^2$.
- Role of W_d (noise) and W_m (model constraints); damping λ .

- Challenges: non-linearity (Jacobian needed), non-uniqueness (multiple m give similar $g(m)$), ill-conditioning.

40–60 min: Activity 3.A: Testing Linearity

Two conditions must be met for linearity. Let's explore how to rigorously test for linearity before we work on other functions.

60–90 min: Activity 3.B: Linear or Non-linear?

Using the equation handout:

- Identify model parameters and classify each relation as *linear*, *non-linear*, or *linearizable*.
- If non-linear, propose a linearization (log-transform, re-parameterization, Taylor expansion/Jacobian).
- Deliverable: brief notes per group.

70–85 min: Activity 3.C: Buried Sphere — Background

We're going to look at the gravity anomaly from a buried sphere. The forward problem is non-linear with respect to the model parameters, requiring a non-linear inverse solution. This activity is broken into two parts: a look at the formulation, and an investigation using a simple program.

- App overview: Data & Model view; Misfit view; controls for acquisition, noise, synthetic model.
- Construct a true model & generate data (freeze a single noise realization).

85–105 min: Buried Sphere — Jacobian Derivatives

1. Students derive/confirm analytical expressions for $\partial g / \partial z$, $\partial g / \partial R$, $\partial g / \partial \Delta \rho$.
2. Compare with the app's Jacobian; discuss relative sensitivities: amplitude $\propto R^3 \Delta \rho$, width controlled by depth z .

105–150 min: Buried Sphere — Investigation

With the use of `buried_sphere_gui.py`, we will examine a number of aspects of inversion, including

Grid search: Explore pairwise misfit slices and 3-D misfit surface. Identify local vs global minima.

Inversion path: Run Gauss–Newton with different λ ; examine step length and stability.

Effect of noise: Increase σ ; observe widening misfit basins and parameter uncertainty.

Non-uniqueness: Demonstrate parameter trade-offs (e.g. $a^3 \Delta \rho$) and resolution ellipses.

Role of Noise: Look at how noise affects results.

150–160 min: Debrief

- Key takeaways: forward/inverse distinctions, misfit and weighting, non-linearity, non-uniqueness, damping.
- Open Q&A; preview of next week's material.

160–170 min: Preview: Gravity Theory

This week we treated gravity primarily as data to be inverted. Next week we will begin developing gravity theory itself.

We will derive the gravity field from Newton's law of universal gravitation and examine how mass distributions create gravity anomalies. We will see how simple geological bodies such as spheres, cylinders, slabs, and contacts produce characteristic gravity signatures.

Along the way, we will connect the mathematical concepts introduced in Week 1—gradients, curvature, and potential fields—to one of the most important geophysical methods.

By the end of next week, you will be able to predict how changes in density, geometry, and observation distance affect the gravity field and begin interpreting gravity anomalies in geological terms.

Materials

- Linear/non-linear handout or slide deck.
- Activity D handouts (gravity and magnetic curves-only plots).
- Buried sphere app (laptop per group).
- Whiteboard or shared notepad for Jacobian derivations.

Assessment

- Activity notes (classification & argumentation).
- Jacobian derivation steps.
- Short reflection or exit-ticket on non-linearity and non-uniqueness.

References (suggested for students)

Further Reading:

- [1] R. Blakely, *Potential Theory in Gravity and Magnetic Applications*, Cambridge University Press, 1995.
- [2] W. M. Telford, L. P. Geldart, and R. E. Sheriff, *Applied Geophysics*, 2nd ed., Cambridge University Press, 1990.
- [3] W. Menke, *Geophysical Data Analysis: Discrete Inverse Theory*, 3rd ed., Academic Press, 2012.
- [4] R. C. Aster, B. Borchers, and C. H. Thurber, *Parameter Estimation and Inverse Problems*, 3rd ed., Elsevier, 2019.
- [5] A. Tarantola, *Inverse Problem Theory and Methods for Model Parameter Estimation*, SIAM, 2005.

Workshop Notes

Workshop Notes

Activity 3.A

Testing Linearity

Purpose

Linear functions can be solved in a single inversion step using linear least squares. This method is simpler, faster, and less computationally expensive, so when possible a linear function is preferred. In this activity, you'll learn to determine whether a function is linear or non-linear.

Learning Objectives

By the end of this activity you should be able to:

- describe the two conditions for linearity, and
- be able to rigorously demonstrate whether a function is linear or non-linear using these tests.

Conditions for Linearity

There are two conditions that must be met for a function to be linear: homogeneity,

$$f(\alpha m_1) = \alpha f(m_1),$$

and additivity

$$f(m_1 + m_2) = f(m_1) + f(m_2).$$

Note: something that may take some getting used to is that our tests for linearity are with respect to model parameters that we are trying to estimate via inversion, not the experimental conditions or observation position, which is how we otherwise normally think of functions.

Box .1: Example of a linear function

For instance, $y = ax + b$ is linear with respect to the model parameters $\mathbf{m} = (a, b)$. For homogeneity,

$$\begin{aligned} y(\alpha \mathbf{m}) &= (\alpha a)x + (\alpha b) \\ &= \alpha(ax + b) \\ &= \alpha y(\mathbf{m}). \end{aligned}$$

For additivity, where $\mathbf{m}_1 = (a_1, b_1)$ and $\mathbf{m}_2 = (a_2, b_2)$,

$$\begin{aligned} y(\mathbf{m}_1 + \mathbf{m}_2) &= (a_1 + a_2)x + (b_1 + b_2) \\ &= (a_1x + b_1) + (a_2x + b_2) \\ &= y(\mathbf{m}_1) + y(\mathbf{m}_2). \end{aligned}$$

Since both homogeneity and additivity are satisfied, the function is linear.

Question 1. Linear function

Show that

$$k(\text{SiO}_2) = a(\text{SiO}_2)^2 + b(\text{SiO}_2) + c$$

is linear with respect to the model parameters, $\mathbf{m} = (a, b, c)$.

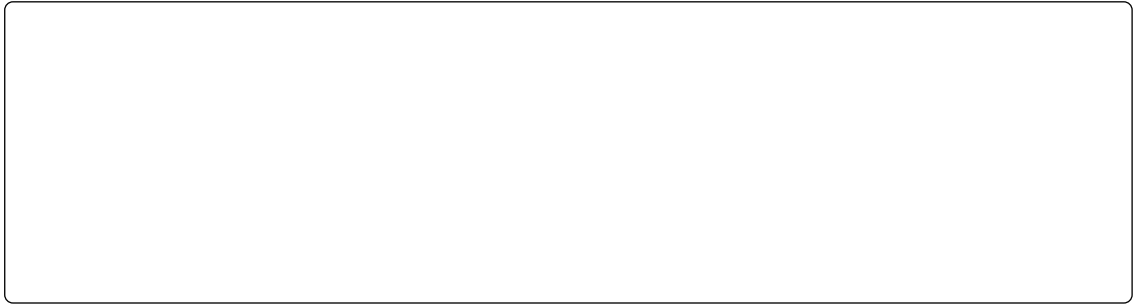
Question 2. Non-linear function

Show that

$$k(T) = k_0 \left(\frac{298}{T} \right)^n$$

is non-linear with respect to the model parameters, $\mathbf{m} = (k_0, n)$.

Question 3. Why is the first equation easier to invert than the second?



Activity 3.B

Linear or Non-linear?

Goal. For each of the following forward relations drawn from potential fields, geothermics, and physical property relations, decide whether it is (i) linear, (ii) non-linear, or (iii) non-linear but can be linearized. If non-linear, describe how you would linearize or re-parameterize it. Justify your choice with respect to the model parameters that would be inverted.

Inversion problems can be cast as $\mathbf{A}\mathbf{m} = \mathbf{d}$ for linear problems and $A(m) = d$ for non-linear problems. Where d is the data and A is the operator or design (more on that below). The difference may appear subtle, though in reality the difference is that the linear equation can be written as a linear algebra problem as the model parameters are independent and discrete. In the non-linear formulation, the one or more model parameters are dependent upon another. As a result they cannot be fully separated unless they can be linearized.

Let's explore what A , m , and d are a bit more intuitively. The data, d , are the results of an experiment. Identifying the data are generally straightforward. The operator and model parameters can take a bit more getting used to. The model parameters, m , are the physical or virtual properties of the system and initially unknown. Once you determine these properties, the forward model becomes a prediction. The operator or design matrix, A , can be thought of *as the experiment itself*, and is dependent upon the geometry, conditions of the experiment, and the physics. The model tells us what exists, the operator tells us how we observe it.

When we are doing a geophysical experiment with the Earth as the medium,

- $\mathbf{m} \rightarrow$ the model is Earth, what it is made of and its physical properties. Changing $\mathbf{m}$ changes the Earth.
- $\mathbf{A} \rightarrow$ how Earth talks to the instrument, how the Earth is sampled by the experiment, i.e., the rules of the experiment. Changing $\mathbf{A}$ changes the experiment, i.e., same Earth + different experiment $\rightarrow$ different data.
- $\mathbf{d} \rightarrow$ the data is the outcome of the experiment, what we measure

In most geophysical problems, $\mathbf{A}$ is

- gravity – distances from mass elements
- magnetics – sensor positions and dip angle
- heat flow – layer thicknesses and boundary conditions
- electromagnetics – distances from conductive/resistive bodies
- seismology – ray paths or travel times

So the entries in $\mathbf{A}$ are determined by:

- positions
- distances
- times
- physical laws
- boundary conditions

Put simply, the matrix $\mathbf{A}$ encodes the physics and geometry of how the experiment samples the model, $\mathbf{m}$, resulting in the observations, $\mathbf{d}$.

Instructions (15–20 min)

1. Work in groups of 3–4.
2. Identify the data and the experimental quantities.
3. For each equation
 - identify the quantity or quantities that would be treated as the *model parameter(s)* in an inversion.
 - Classify the mapping data $\rightarrow$ model as linear or non-linear in those parameters.
 - If non-linear, propose a linearization (e.g., log-transform, Taylor expansion/Jacobian around p_0 , or a change of variables) and note any loss of identifiability.

Equations

- Q1. **Gravity (Bouguer slab):** We'll see this equation again as it is used to model the gravity due to an infinite slab.

$$\Delta g = 2\pi G(\rho_s - \rho_c)H$$

- Q2. **Gravity (buried sphere):** This gravity solution is one of the simplest models for a gravity anomaly. We will explore it further.

$$g(r) = \frac{4}{3}\pi a^3 G(\rho_s - \rho_c) \frac{z}{(r^2 + z^2)^{3/2}}$$

- Q3. **Thermal conductivity $k(T, X)$:** We'll use this formulation in the problem set. This equation is used to estimate thermal conductivity model for a solid solution series for a mineral.

$$k(T, X) = (a + bX + cX^2) (298/T)^n$$

- Q4. **Thermal conductivity $k(T, P)$:** This form of thermal conductivity is often used to estimate thermal conductivity for a rock.

$$k(T, P) = \frac{k_0}{A + BT} (1 + cP)$$

- Q5. **Isostasy (Airy):** Airy isostasy is an end-member isostatic model, used to explain topographic variations due to differences in crustal thickness.

$$\Delta t = \frac{\rho_c}{\rho_m - \rho_c} h$$

- Q6. **Isostatic gravity correction (column):** Gravity can be corrected for a mass deficit created by variations in crustal thickness via an isostatic correction.

$$\Delta g_{\text{iso}} \approx 2\pi G(\rho_m - \rho_c) \Delta t - 2\pi G \rho_c h$$

- Q7. **Magnetics (2-D infinite dike):** The ocean floor records chaotic reversals in Earth's magnetic field. The remnant magnetic field is "frozen" in dikes produced at the mid-ocean ridge.

$$\Delta T(x) = C \left[\arctan\left(\frac{x - x_1}{z}\right) - \arctan\left(\frac{x - x_2}{z}\right) \right]$$

- Q8. **Layered 1-D heat with sources:** Geotherms can largely be explained by 1-D heat flow through a layered Earth. We'll see this again when we start geothermics.

$$T_i(z) = T_{i-1} + \frac{q_{i-1}}{k_i}(z_i - z_{i-1}) - \frac{A_i}{2k_i}(z_i - z_{i-1})^2$$

- Q9. **Density-porosity mix:** Sedimentary rocks include pore space (porosity) that may be filled with a fluid (air, water, brine, gas, oil, etc.).

$$\rho_{\text{bulk}} = (1 - \phi) \rho_{\text{matrix}} + \phi \rho_{\text{fluid}}$$

- Q10. **Gardner's relation (density-velocity):** Many solutions in geophysics involve trying to estimate one property from another, in this case density from seismic velocity.

$$\rho = a v^b$$

- Q11. **Gravity (discrete 2-D prisms):** While we can produce analytic equations for a variety of idealized geologic bodies with simple geometries (e.g., buried spheres as above), to produce more geologically

realistic bodies we can produce structures from arbitrarily-shaped polygons.

$$g_z = 2G\Delta\rho \sum_{i=1}^N \left[(x_{i+1} - x_i) \log\left(\frac{r_{i+1}}{r_i}\right) + (z_{i+1} - z_i) \arctan\left(\frac{x_i z_{i+1} - x_{i+1} z_i}{x_i x_{i+1} + z_i z_{i+1}}\right) \right]$$

where

$$r_i = \sqrt{x_i^2 + z_i^2}, \quad (x_{N+1}, z_{N+1}) = (x_1, z_1).$$



Activity 3.C

Inversion of a Buried Sphere

Purpose

The purpose of this activity is to develop an intuitive understanding of inversion using the gravity anomaly of a buried sphere. By comparing grid-search and gradient-based approaches, you will explore how inverse methods search model space, how model parameterization influences the solution, and why non-uniqueness and noise make geophysical interpretation challenging.

Learning Outcomes

By the end of this activity, you should be able to:

- Explain the difference between forward modelling, grid search, and gradient-based inversion.
- Describe the role of the Jacobian matrix in determining model updates.
- Identify sources of non-uniqueness in geophysical inverse problems.
- Explain how model parameterization can improve or degrade inversion performance.
- Describe the influence of noise on inversion results and uncertainty.

Gravity of a Buried Sphere

We will return to how we derive the gravity due to a buried sphere later. For now, we will model the vertical gravity anomaly at the surface produced by a buried sphere of depth z (to center), radius a , and density contrast $\Delta\rho$. For a station at horizontal radius $r = \sqrt{x^2 + y^2}$,

$$g(r|z_c, a, \Delta\rho) = \underbrace{\frac{4}{3}\pi a^3 \Delta\rho}_{\text{mass}} G \frac{z}{(r^2 + z_c^2)^{3/2}}, \quad (8)$$

with G the gravitational constant.

Inverse Model

The gravity equation due to a buried sphere is non-linear, requiring a non-linear iterative solver. The code we will use employs the Gauss–Newton inverse method, which can be enhanced with Levenberg–Marquardt regularization.

We seek $\mathbf{m} = [z_c, a, \Delta\rho]^T$ that minimizes the objective function,

$$\Phi(m) = \underbrace{\|W_d(\mathbf{g}_{\text{obs}} - g(\mathbf{m}))\|_2^2}_{\text{model misfit}} + \underbrace{\lambda^2 \|\mathbf{m} - \mathbf{m}_0\|_2^2}_{\text{regularization term}},$$

where $\mathbf{g}_{\text{obs}}$ are the observed gravity data, $g(\mathbf{m})$ is the predicted gravity using Equation 8, $\mathbf{W}_d$ is a diagonal matrix of inverse variance (data uncertainty), and λ is the Levenberg–Marquardt regularization parameter. Our initial guess is given by the column vector $\mathbf{m}_0$.

To determine $\mathbf{m}$, we have to solve iteratively, with each model step being

$$\Delta\mathbf{m} = (\mathbf{J}^T \mathbf{W}_d^2 \mathbf{J} + \lambda^2 \mathbf{I})^{-1} \mathbf{J}^T \mathbf{W}_d^2 (\mathbf{g}_{\text{obs}} - g(\mathbf{m}_0)).$$

Role of λ . Small λ approaches Gauss–Newton (fast, but can be unstable if $J^\top J$ is ill-conditioned). Large λ yields smaller, more stable steps (gradient-descent-like), helpful far from the solution or with strong trade-offs.

Question 1. Jacobian

Before we play with the buried sphere program, let's derive derivatives we need for the Jacobian matrix.

The Jacobian for the buried sphere requires three derivatives:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial g}{\partial z_c} & \frac{\partial g}{\partial a} & \frac{\partial g}{\partial \Delta\rho} \end{bmatrix}.$$

Derive the derivatives from Equation 8.

Exploration

The questions below relate to the `buried_sphere_gui.py` program. While the questions are about the demo, the answers are largely independent of both the problem and program.

Question 2. Grid Search

What is a grid search? Is it a forward or inverse driven solution? What does it tell us about the misfit? How do we determine the model parameters?

Question 3. Inversion

How does inversion approach the problem differently from the grid search? What path does it take to the answer?

Question 4. Comparison

How does the result of the inversion compare with that of the grid search? If the two solutions do not match, why?

Question 5. Non-uniqueness

The problem we have posed is non-unique even though we use a single "true" model to generate the data. Why is the solution non-unique?

Question 6. 2D versus 3D

Let's switch from 3 parameters to 2. The 2 parameter result is a better constrained, why? How is this improvement in constraint reflected in the misfit surfaces?

Question 7. Noise

How does noise affect the solution via inversion? How does noise contribute to non-uniqueness?

For this demonstration, repeat the inversion with 0%, 5%, and 10% Gaussian noise.

Key Ideas

Grid search is a brute-force forward modelling approach to systematically sample the model space with misfit is computed at every combination of model parameters. It can identify the global minimum within the explored model domain, but it becomes computationally expensive as the number of model parameters increases and requires a sufficiently dense sampling of the model space to accurately resolve the misfit surface. For many large difficult problems found in geophysics, a grid search is unfeasible.

Inversion uses information about the local shape of the misfit surface, typically through the Jacobian matrix, to move iteratively toward a low-misfit solution. While each iteration is computationally more expensive than a single forward calculation, inversion generally requires far fewer model evaluations than a grid search.

Best of all, inversion remains practical for geophysical problems with many parameters (tens to millions). The trade-off is that inversion explores only a small portion of model space and may converge to a local minimum rather than the global minimum.

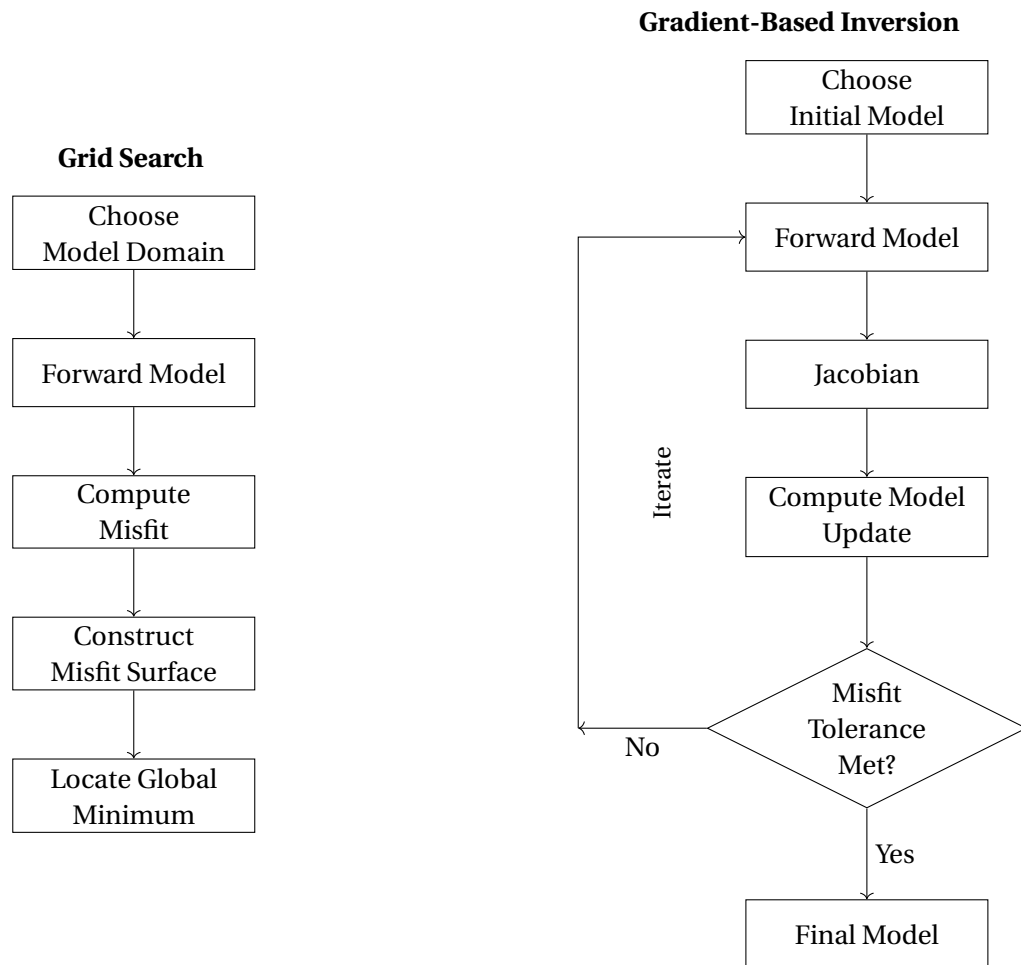


Figure 7: Comparison of grid-search and gradient-based inversion approaches.

Student Name: _____

Student ID: _____

Instructions (Instructor Use)

Circle the level that best reflects the student's overall performance since the previous checkpoint. Use the booklet as supporting evidence, but prioritise observed participation.

Criterion	HD	D	C	P	F
Preparation & Contribution	Insightful, advances discussion	Regular, clear reasoning	Occasional, appropriate	Basic, prompted only	Little or none
Engagement with Others	Synthesises ideas, elevates group	Builds on others' ideas	Responds to peers	Listens, limited interaction	Disengaged
Curiosity & Thinking	Explores ambiguity, asks deep questions	Engages with complexity	Asks genuine questions	Attempts answers when prompted	Avoids engagement
Booklet Evidence	Complete, thoughtful, annotated	Mostly complete, good detail	Adequate, some gaps	Minimal effort evident	Sparse or missing

Overall Performance

HD

D

C

P

F

Notes (optional)

Week 4 — Gravity: Density, Structures, and Non-Uniqueness

Duration: 3 hours

Setting: Workshop room (whiteboards; projector)

Preparation

Pre-reading: Course Notes, Ch. 6

Lecture videos:

Notes:

Purpose

This workshop develops intuition about how density varies in porous and compacted materials, how different subsurface structures relate to density contrasts, and how the buried sphere solution illustrates non-uniqueness in gravity. Students work from physical reasoning, analytic solutions, and qualitative cross-section analysis. The session prepares students for Week 5 (filters, upward/downward continuation, reduction-to-the-pole).

Learning Outcomes

- Estimate bulk density of porous materials using simple mixing laws.
- Explain how compaction and pore-fluid substitution change density.
- Predict density contrasts for various geologic structures.
- Interpret gravity anomaly width–depth relationships for the buried sphere.
- Recognise non-uniqueness and explain why lenticular shallow bodies can mimic deep spheres.

Session Flow (170 min)

Mixed groups of 4–5 with geology and physics/engineering backgrounds.

0–15 min: Warm-up — What does gravity see?

In groups, rank a series of geological changes according to the density contrasts they are likely to produce. Consider examples such as felsic versus mafic rocks, porous versus compacted sediments, and common metamorphic transformations. Discuss which changes would be most visible to a gravity survey and which might produce similar gravity responses despite having very different geological origins.

Key idea: Gravity does not measure rock names or geologic processes directly—it responds to density contrasts. Different geological processes can therefore produce similar gravity anomalies, introducing non-uniqueness into interpretation.

15–30 min: Mini-lecture

30–70 min: Activity 4.A: Density of a Two-Component Porous Sandstone

Examine how porosity evolves during burial and how compaction changes bulk density. Using real porosity–depth data, develop a simple compaction model and explore how the resulting density changes influence gravity interpretation.

70–110 min: Activity 4.B: Geologic Structures to Gravity Anomalies

In this activity, you will discuss in groups some basic structures and how they connect to geology and density contrast. You will then make an educated guess what the gravity anomaly in response to the structure will be.

110–150 min: Activity 4.C: Buried Sphere — Width to Depth

Work with the buried sphere analytic solution to explore non-uniqueness.

Explain why increasing depth increases width but decreasing $\Delta\rho$ does not. Discuss non-uniqueness: why can shallow, lens-shaped bodies (“lenticular”) mimic deep spheres?

150–160 min: Wrap-up

1. Which factor (porosity, fluid, composition) most strongly affects density in shallow basins?
2. Which structural density contrast is hardest to distinguish with gravity alone?

160–170 min: Preview: Isostasy

This week we explored how density contrasts produce gravity anomalies. A natural next question is whether large topographic features such as mountain belts should therefore always produce large positive gravity anomalies. Surprisingly, they often do not. Next week we will investigate isostasy—the idea that columns of rock tend toward gravitational balance. We will compare the structure of oceanic and continental lithosphere, examine why topography requires compensating mass variations at depth, and derive the isostatic gravity correction. Along the way, we will see why crustal thickness, mantle density, and lithospheric structure are all important for understanding Earth’s large-scale gravity field.

Workshop Notes

Workshop Notes

Activity 4.A

Density and Porosity

Purpose

To develop intuition for how porosity affects rock density and how compaction during burial modifies density contrasts relevant to gravity interpretation. By examining real porosity–depth data, students will identify simple mathematical descriptions of compaction and explore how changes in porosity influence the bulk density of sedimentary rocks.

Learning Outcomes

By the end of this activity, you should be able to:

- describe how porosity varies with burial depth and explain the role of compaction,
- identify a plausible mathematical model for porosity reduction with depth and interpret its parameters,
- derive and apply a bulk-density relationship for porous rocks,
- explain how compaction alters density contrasts between sedimentary rocks and crystalline basement,
- relate density changes caused by porosity loss to qualitative gravity responses.

Introduction

In week 2, we calculated several physical properties, though in each case we assumed porosity was negligible. However for many rocks, especially sedimentary rocks near the surface, porosity can be significant.

What is porosity? Porosity is void space, but that doesn't mean it is empty. Other than an unsaturated zone at the surface the voids are generally saturated with fluid (water, brine, steam, oil, gas, H_2). The fraction in sediments can be 30% for porous sandstone up to ~70% at the seafloor. Though for most sedimentary rocks it is <20% and crystalline—plutonic and metamorphic rocks—porosity is often $\lesssim 2\%$. Porosity comes in a few flavors:

intergranular – between mineral grains, common in sedimentary and volcanic rocks,

vesicular – pore space created by gas bubbles, common in volcanic rocks, particularly shallow erupted basalts,

fracture – cracks and joints created by tectonic activity, decompression near the surface, cooling, or meteorite impacts (minor),

weathering related – generally dissolution by groundwater.

Most porosity loss occurs within the upper ~10–11 km of the crust, where increasing overburden stress closes fractures and compacts sediments. Small amounts of porosity may persist to greater depths.

Porosity is generally reported as a decimal fraction between 0 and 1, or as a percentage of the total volume of the rock. Only porosity will be reported, not the solid component (matrix), so what fraction makes up the this portion?—well the rest of it of course!

Tasks

In the figure below you will see sediment porosity–depth data for a chalk section off the coast of Norway. Estimate the nature of the compaction.

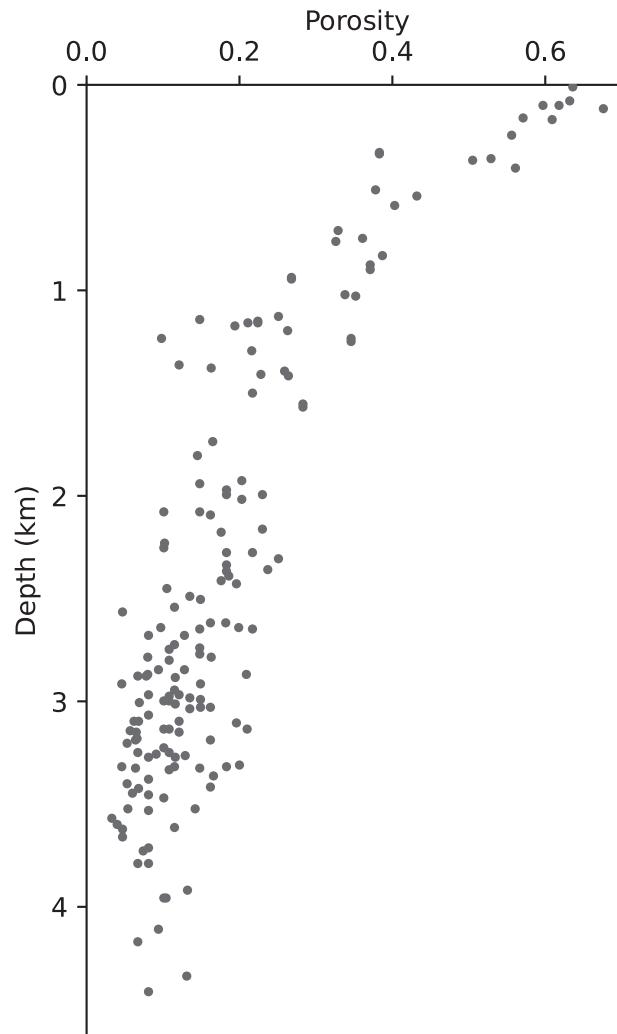


Figure 8: Porosity data from Mallon and Swarbrick (2002) on calcareous ooze.

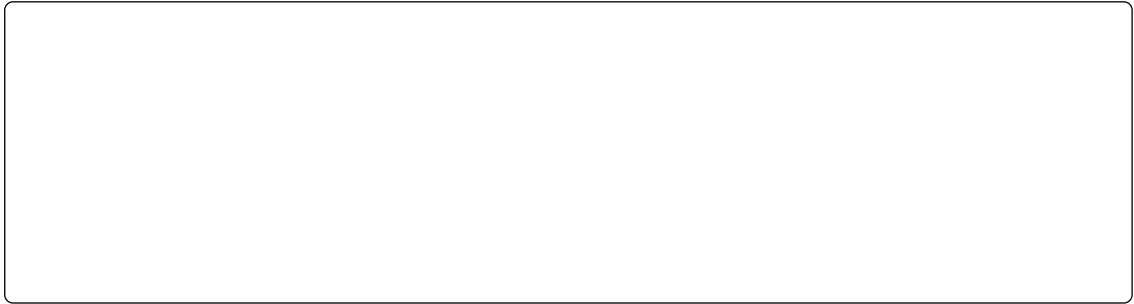
Question 1. Compaction of Sediments

To estimate the compaction relationship with depth, start by drawing an approximate curve for porosity from the surface to 4 km depth.

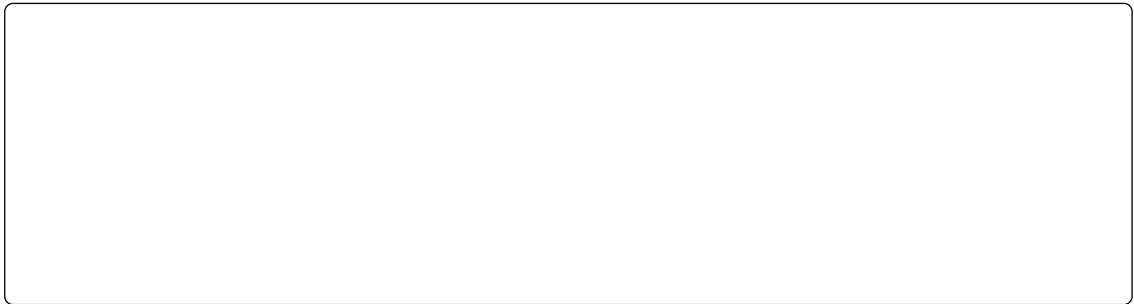
Which of the mathematical models appears to fit the data best? And, why do all of these models include a z/h term?

a. $\phi(z) = \phi_0 - zh^{-1}$ b. $\phi(z) = \phi_0 \exp(-zh^{-1})$ c. $\phi(z) = \frac{\phi_0}{1 + zh^{-1}}$

Assuming your preferred model, estimate values for ϕ_0 and h ? Also label these on the figure on the appropriate axes.



What physical meaning would you assign to: (a) ϕ_0 , (b) h ?



Question 2. Density of Sediments

If the density of the matrix material, ρ_m , in the chalk examined above ranges from 2650 to 2720 kg m⁻³, derive a relationship for the bulk density of the sediment with pores filled with seawater ($\rho_f = 1030$ kg m⁻³).



Which depth interval produces the largest change in density: 0–1 km, 1–2 km, 2–3 km, or 3–4 km? Why?



What is the density of the sediments at the surface and h ? What is the relative change in density?



Question 3. From Porosity to Gravity

As sediments become buried and compacted, do you expect the density contrast between the sediments and crystalline basement to increase or decrease?

What implications does this have for gravity interpretation?

Activity 4.B

Geologic Structures to Gravity Anomalies

Purpose

To develop intuition for how geological structures are represented by idealized geometric bodies in gravity interpretation. By relating common geologic features to simple analytical models, you will explore how density contrast, geometry, and burial depth influence the character of gravity anomalies.

Learning Outcomes

By the end of this activity, you should be able to:

- identify geological structures that can be approximated by common gravity-model geometries,
- estimate plausible density contrasts for a variety of lithologies and geologic settings,
- predict the qualitative shape of gravity anomalies from simple subsurface structures,
- relate anomaly symmetry, width, amplitude, and gradients to source geometry,
- explain the advantages and limitations of using simplified geometric models in gravity interpretation.

Introduction

The structural arrangement of rock units can be very complex. As geophysicists we invert for more complex structures as necessary, but doing so is time-consuming. We can often approximate complex structures by one or more simplified geometries. These simplifications have benefits besides being faster. For one, the models have closed form analytical solutions. We can calculate a wide range of structures that vary in dimension, distance, or orientation rapidly giving us approximate solutions that in many cases are 'good enough' given all the data uncertainties and model non-uniqueness issues. Another advantage is that closed form solutions help build our intuition in a way that inverting for cell properties do not.

For each structure below, complete the table and use the *upper axes* to sketch the expected gravity anomaly $g_z(x)$ (arbitrary units). The *lower panel* shows a schematic cross-section (z positive downward).

In your groups:

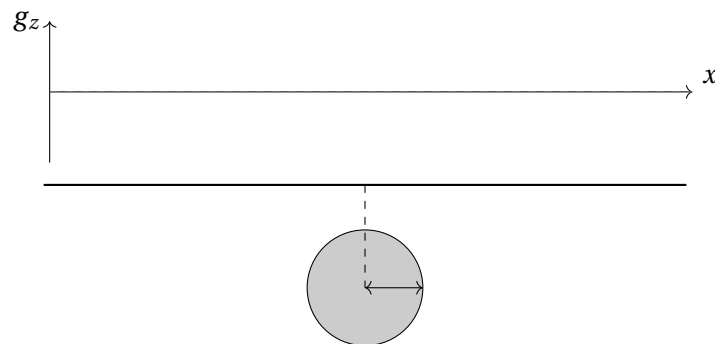
1. Discuss geologic structures that may be described by the simple shapes.
2. Propose plausible density contrasts ($\Delta\rho$) given typical lithologies.
3. Predict qualitative anomaly shape: narrow/broad? symmetric/asymmetric? positive/negative? asymptotic limit?
4. One group presents each structure; whole class discusses similarities and differences.

Table

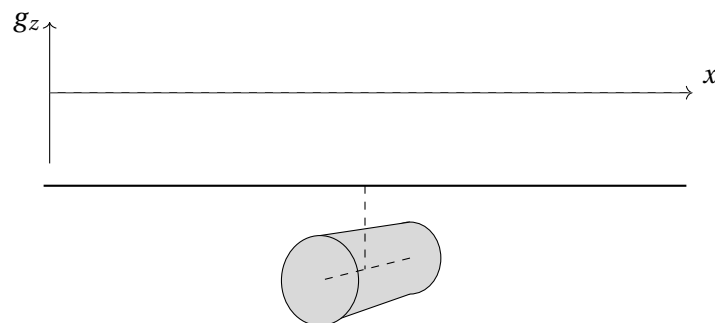
	Geologic Structure(s)	Likely lithologies	Plausible Range $\Delta\rho$ (kg m^{-3})
A.			
B.			
C.			
D.			
E.			
F.			
G.			

Prompts: Indicate sign (pos./neg.), width (narrow/broad), symmetry/asymmetry, and where the steepest gradients occur.

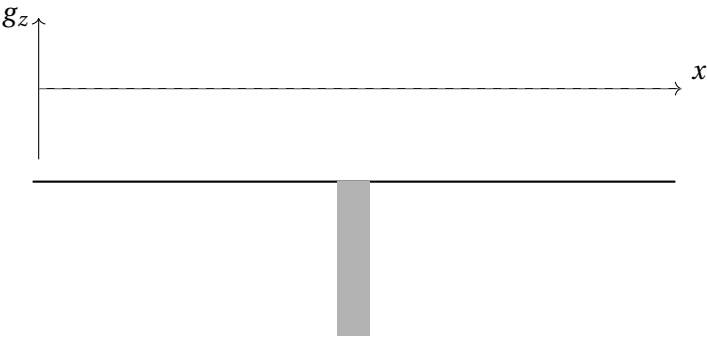
A) Buried Sphere (3D)



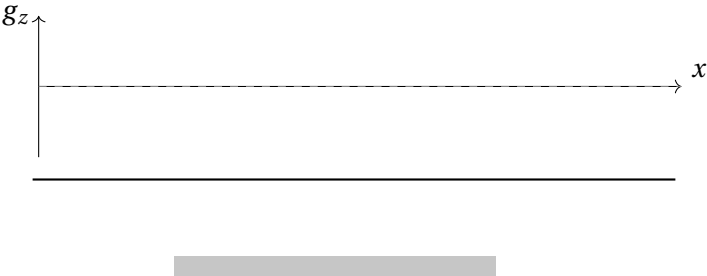
B) Buried Cylinder (2D; infinite strike)



C) Vertical Slab



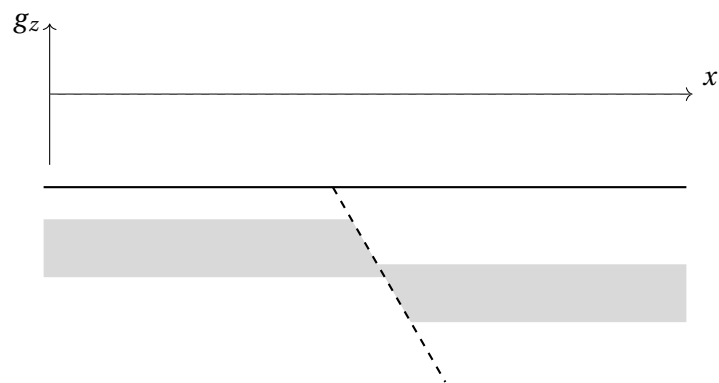
D) Finite-Extent Horizontal Slab



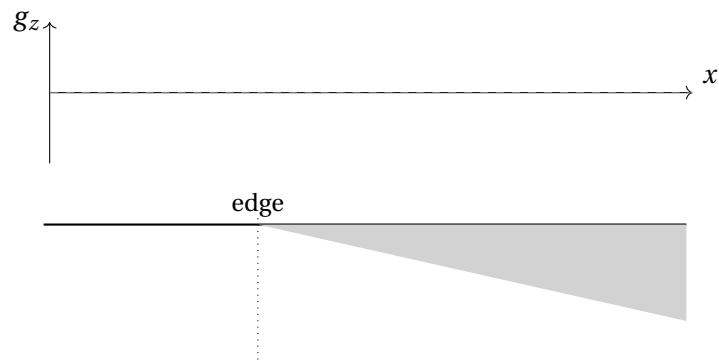
E) Inclined Plane



F) Offset Layer



G) Gently Sloped Basin (edge at one-third from left boundary)



Activity 4.C

Buried Sphere — Width to Depth

The gravity anomaly due to a buried sphere is given by:

$$g_z(x) = \frac{4}{3}\pi a^3 G \Delta\rho \frac{z_c}{(x^2 + z_c^2)^{3/2}},$$

where a is the radius of the sphere, $\Delta\rho$ is its density anomaly, and z_c is the depth to the center of the sphere. The coordinate x is the position along the profile at Earth's surface.

Last week, we saw this relationship leads to non-uniqueness because $a^3 \Delta\rho$ are inseparable. Now, however, we want to show that one can produce a simple relationship between the depth of the buried sphere and the width of its anomaly, independent of the mass. Such a relationship is useful for estimating the depth to structures that produce peaks in the gravity field. Such a relationship is useful for producing back of the envelope estimates in the field or from a map before any formal modeling is done. Such relationships are the prelude concepts that lead to wavelength filtering that we will see in Week 7.

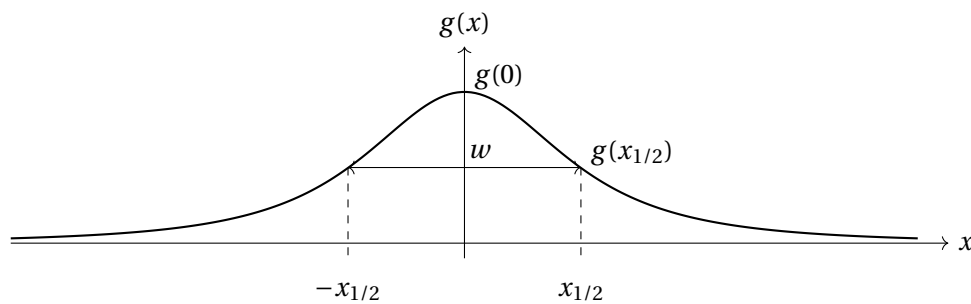


Figure 9: Gravity anomaly due to a buried sphere.

Tasks

To develop the depth to width relationship:

Question 1. Peak Anomaly

Show that $g_z(0)$ is proportional to $\Delta\rho a^3 / z_c^2$.

Question 2. Width to Depth Relationship

Show that the depth

$$z_c \approx 0.65 w$$

where w is the full-width half-maximum.

To do so, solve for $x_{1/2}$ starting from $g_z(x_{1/2}) = \frac{1}{2} g_z(0)$. Hint: this is on the right track,

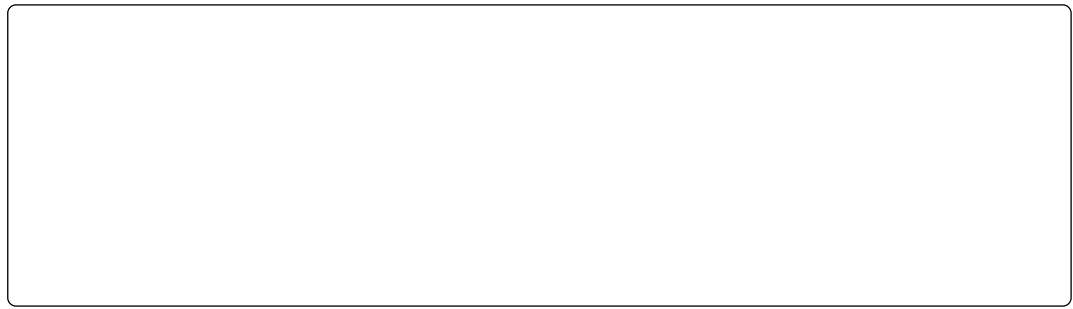
$$\frac{z_c}{x_{1/2}} = (2^{2/3} - 1)^{-1/2}.$$

Question 3. Interpretation

- a Why does depth affect width but density contrast does not?

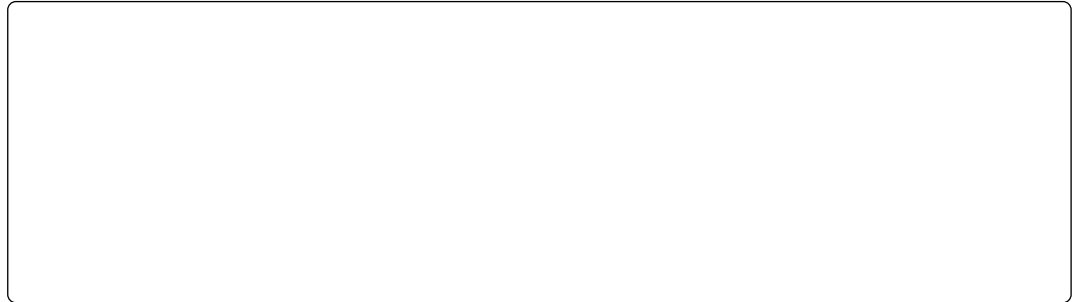
36pt

- b How do the radius and density contrast result create non-uniqueness for inversion and how we ensure uniqueness?



36pt

- c Why does a shallow lenticular body (e.g., laccolith or lopolith) could mimic a deep sphere's anomaly?



36pt

Week 5 — Isostasy: Ocean–Continent Structure & Isostatic Gravity

Duration: 3 hours

Setting: Workshop room (small-group whiteboard work)

Preparation

Pre-reading: Course Notes, Ch. 7

Lecture videos:

Notes:

Purpose

This workshop develops a quantitative understanding of isostatic equilibrium and its gravity expression by combining column-based reasoning, ocean–continent structure, and slab gravity approximations. Airy and Pratt isostasy are treated as end-member models, with emphasis on why real Earth structure requires a mixed or layered approach involving the lithospheric mantle.

Learning Outcomes

By the end of this workshop, you should be able to:

- Explain isostatic equilibrium as equal lithostatic pressure (or mass per unit area) at a compensation depth.
- Derive and interpret Airy and Pratt isostasy as conceptual end members.
- Explain why crust-only isostatic models fail for ocean–continent systems.
- Derive the Airy isostatic gravity correction using slab approximations.
- Explain why gravity anomalies over high topography may be small.

Session Flow (180 min)

0–10 min: Warm-Up: Defining Isostasy

Individually, write a one-sentence definition of isostatic equilibrium. These definitions are compared in pairs and then within your group, with the goal of converging on a precise statement.

Target idea:

Isostatic equilibrium requires equal lithostatic pressure (or mass per unit area) at a compensation depth beneath adjacent columns.

Groups assign roles (facilitator, skeptic, recorder, assumption checker).

10–30 min: Mini-Lecture: Airy, Pratt, and Mixed Isostasy

A short instructor-led lecture reviews:

- Airy isostasy: variable crustal thickness, uniform density
- Pratt isostasy: constant thickness, lateral density variations
- Why neither end member fully describes the Earth
- The role of lithospheric mantle and partial compensation

This provides the conceptual framework for the two analytical activities that follow.

30–70 min: Activity 5.A: Oceanic vs Continental Columns

Purpose

To test whether crust-only Airy isostasy can account for observed ocean–continent elevation differences, and to explore how including the lithospheric mantle resolves inconsistencies.

Crust-Only Airy Isostasy

Each group balances an oceanic and a continental column under the assumption that compensation occurs solely through crustal thickness. You will:

- Enforce equal mass per unit area above a compensation depth.
- Solve for the oceanic crust density required for equilibrium.
- Compare your result with realistic oceanic crust densities.

You should find that the required density is unrealistic, indicating a missing component in the model.

Two-Layer Lithosphere Extension

The model is then extended to include a lithospheric mantle beneath both oceanic and continental crust. You will:

- Rebalance the columns including mantle lithosphere thickness differences.
- Estimate the density contrast between oceanic and continental lithospheric mantle.
- Relate this contrast to mineralogical differences (e.g. garnet content).

70–80 min: Whole-Class Synthesis

A brief instructor-led discussion synthesises results from Activity 1, focusing on:

- Why crust-only Airy isostasy fails
- Why the lithospheric mantle must be included
- Where Pratt-like behaviour may arise

80–120 min: Activity 5.B: Isostatic Gravity Correction

Purpose

To derive the Airy isostatic gravity correction and explain why large topography does not necessarily produce large gravity anomalies.

Airy Mass Balance

You will show that, under Airy isostasy,

$$\rho_c h = (\rho_m - \rho_c) r,$$

and derive an expression for the crustal root thickness $r(x)$.

Infinite Slab Gravity

Using the slab approximation,

$$\Delta g = 2\pi G \rho t,$$

you will compute the gravitational effects of surface topography and of the compensating crustal root.

Cancellation

By substitution, you will demonstrate that under perfect Airy compensation, the gravity effects of topography and root cancel exactly, and discuss what assumptions are required for this result.

120–145 min: Numerical Demonstration

The instructor presents a live Python demonstration illustrating:

- Airy compensation with varying compensation fractions
- Pratt-style lateral density variations
- Resulting gravity anomalies and residuals

Interpretation of results is emphasised rather than coding details.

145–170 min: Group Synthesis

Each group prepares a brief summary addressing:

- One reason crust-only Airy isostasy fails.
- One way lithospheric mantle structure resolves this failure.
- One equation linking topography, density, and gravity.
- One observation that could reduce non-uniqueness in isostatic interpretation.

170–180 min: Exit Questions

Individually, respond to the following:

1. Identify one unrealistic assumption in either Airy or Pratt isostasy and explain why it fails.
2. Explain how geological or geophysical constraints (e.g. seismology or petrology) help resolve ambiguity in isostatic models.

Workshop Notes

Workshop Notes

Activity 5.A

Isostasy Problem: Oceanic vs Continental Crust

Part A: Crust-only Airy Isostasy

Given:

oceanic crust thickness: $h_{oc} = 7 \text{ km}$

ocean depth: $h_w = 3.7 \text{ km}$

seawater density: $\rho_w = 1030 \text{ kg m}^{-3}$

continental crust thickness: $h_{cc} = 40 \text{ km}$

continental crust density: $\rho_{cc} = 2850 \text{ kg m}^{-3}$

continental elevation: $\varepsilon = 0.8 \text{ km}$

mantle density: $\rho_a = 3330 \text{ kg m}^{-3}$

Instructions: Assume isostatic compensation occurs solely through crustal thickness. Compute the density of the oceanic crust, ρ_{oc} , that balances the oceanic and continental columns. Compare your result with typical oceanic crust densities ($\sim 2900\text{--}3000 \text{ kg m}^{-3}$) and discuss why the crust-only assumption may fail.

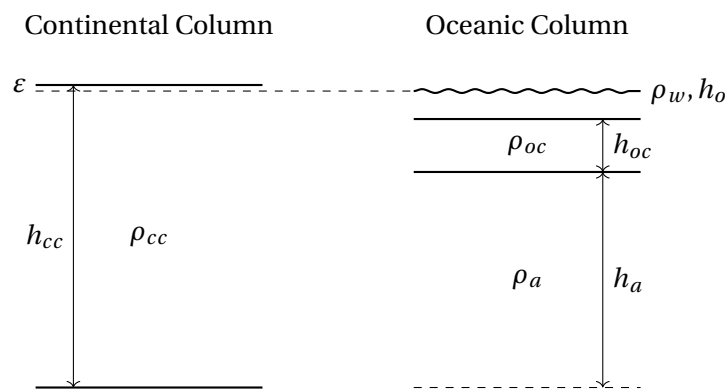


Figure 10: Comparison of isostatic columns for the continental and oceanic crust for Part A.

Questions:

1. Compute the required oceanic crust density ρ_{oc} .
2. Compare with expected oceanic crust densities and discuss why the simple crust-only model may not yield realistic values.

Part B: Two-layer Lithosphere Model

Given additional information:

Oceanic lithospheric mantle thickness $h_{om} = 75 \text{ km}$

Oceanic lithospheric mantle density: $\rho_{om} = 3330 \text{ kg m}^{-3}$

Continental lithospheric mantle thickness: $h_{cm} = 150 \text{ km}$

Continental lithospheric mantle density: $\rho_{cm} = ?$

Asthenosphere density: $\rho_a = 3320 \text{ kg m}^{-3}$

Instructions: Let's add a two-layered lithosphere using average lithospheric thicknesses for the oceanic and continental settings. Our goal will be to see what we can learn about the relative density difference between continental and oceanic mantle lithosphere. In the process, we'll show how including the mantle lithosphere resolves the unrealistic crust-only result.

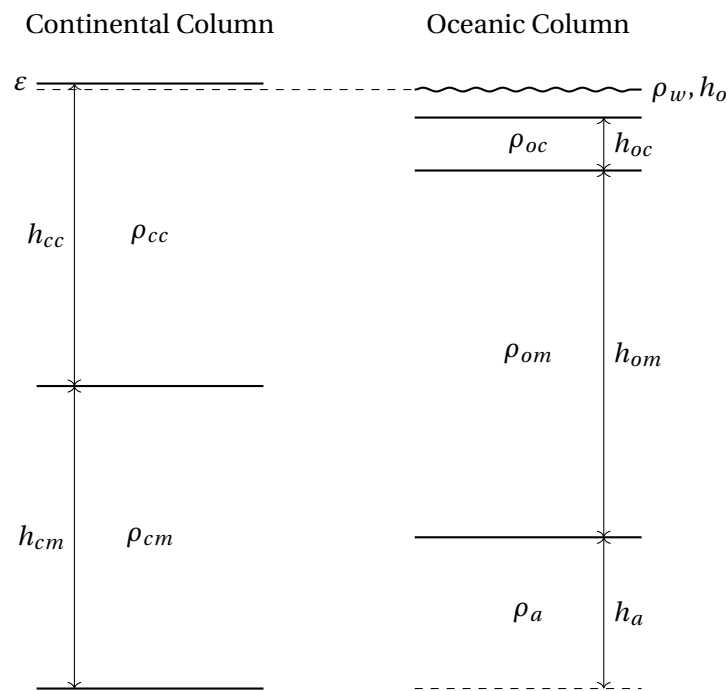


Figure 11: Comparison of isostatic columns for the continental and oceanic crust for Part B.

Table 2: Modal proportions of minerals in the oceanic and continental lithospheres.

Mineral	Oceanic	Continental
olivine	75	60
orthopyroxene	25	17
clinopyroxene	10	11
garnet	0	12

Oceanic lithospheric mantle is assumed to be composed of depleted peridotite with a density of $\rho_{om} = 3330 \text{ kg m}^{-3}$. Continental lithospheric mantle may contain a higher proportion of garnet.

Density of depleted mantle (peridotite): $\rho_d = 3330 \text{ kg m}^{-3}$

Density of garnet: $\rho_g = 3600 \text{ kg m}^{-3}$

Questions:

1. Setup the isostatic equations for a two-layered mantle lithosphere. If we assume an average oceanic crustal density $\rho_{oc} = 2900\text{--}3000 \text{ kg m}^{-3}$, what density of the continental mantle lithosphere?
2. Explain how including the mantle lithosphere resolves the unrealistic density obtained in the crust-only case.
3. Discuss the role of garnet fraction in the continental lithospheric mantle for the density contrast. Compute the percentage difference garnet between the oceanic and continental lithosphere and compare your result with the modal proportions in Table 2.

Activity 5.B

Isostatic Gravity Correction

Purpose

In this problem, you will derive the isostatic gravity correction under the assumption of Airy isostasy, starting from a Bouguer slab formulation and progressing to a practical implementation using topography and a reference crustal thickness.

Background

Consider a reference Earth model with a constant crustal thickness T_0 . Surface topography $h(x)$ is compensated by a crustal root $r(x)$ such that columns of equal width have equal mass above a compensation depth.

Let:

- ρ_c = crustal density
- ρ_m = mantle density
- $h(x)$ = surface elevation
- $r(x)$ = additional root thickness relative to T_0

Geometry

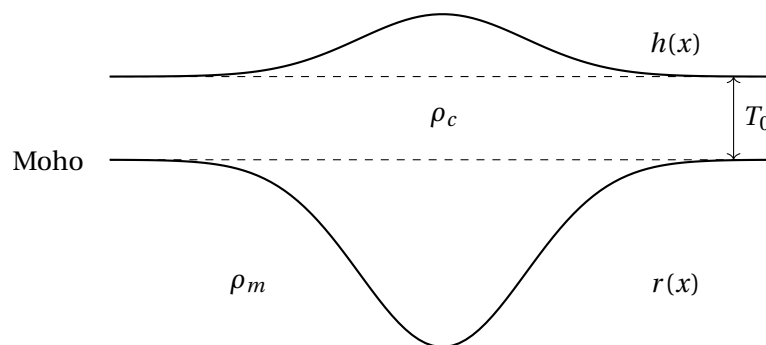


Figure 12: A mountain with a topographic expression $h(x)$ and compensating root $r(x)$.

Part A: Airy Isostasy

1. Starting from mass balance between a reference column and a column with topography, show that:

$$\rho_c h = (\rho_m - \rho_c) r$$

2. Solve for the root thickness $r(x)$ in terms of $h(x)$.

Part B: Gravity of an Infinite Slab

The gravitational attraction of an infinite horizontal slab of thickness t and density ρ is:

$$g = 2\pi G \rho t$$

3. Write an expression for the gravitational effect of the topography.
4. Write an expression for the gravitational effect of the crustal root. Carefully define the density contrast.

Part C: Substitution and Simplification

5. Substitute your expression for $r(x)$ into the root gravity term.
6. Show that the total gravity effect of the topography plus its Airy root simplifies to:

$$g_{\text{total}} = 0$$

Part D: Practical Isostatic Correction

In practice, crustal thickness is not known everywhere. Instead, we assume a reference thickness T_0 and compute $r(x)$ from observed topography.

7. Explain why the isostatic correction is model-dependent.
8. Write the final expression for the isostatic correction in terms of $h(x)$ only.

Target Result

You should show that the gravity effect of the compensating root is:

$$g_{\text{root}} = -2\pi G\rho_c h$$

and that, under perfect Airy compensation, the combined effect of topography and root cancels exactly.

Extension (Optional)

Discuss how this result would change if:

- compensation were incomplete
- the lithosphere behaved elastically (flexural isostasy)

Week 6 — Magnetism: Anomalies and Susceptibility

Duration: 3 hours

Setting: Workshop room (disc magnets, iron filings, susceptibility meter)

Preparation

Pre-reading: Course Notes, Ch. 8

Lecture videos:

Notes:

Purpose

This workshop builds physical intuition for magnetic fields and their geological sources. Using dipole demonstrations, rock susceptibility measurements, and a case study from Antarctica, students contrast magnetism with gravity (vector nature, polarity, temperature dependence) and connect magnetic structure to crustal composition and thermal state.

Learning Outcomes

By the end of this workshop, students should be able to:

- Visualise magnetic fields generated by dipoles and combinations of sources
- Relate magnetic susceptibility to rock type and geological process
- Predict magnetic anomaly shapes from simple geological geometries
- Contrast magnetic anomalies with gravity anomalies for the same structures
- Explain the concept of Curie depth and its relationship to thermal structure
- Link source depth to anomaly wavelength and spectral content

Session Flow (180 min)

0–10 min: Introduction and framing

- How magnetism differs from gravity (vector nature, polarity, temperature dependence)
- Overview of the three activities and how they connect

10–35 min: Activity 6.A: Magnetic fields from dipoles

- Iron filings and disc magnets
- Vertical vs horizontal dipole orientation
- Stacked magnets and multiple sources

Learning emphasis:

- Dipole field geometry
- Orientation-dependent symmetry and polarity
- Intuition for induced magnetisation

35–55 min: Activity 6.B: Whole-class demo: Magnetic susceptibility of rocks

- Susceptibility measurements of igneous, metamorphic, and sedimentary rocks
- Effects of alteration, metamorphism, and weathering

Learning emphasis:

- Susceptibility as a material property
- Why similar compositions can have very different magnetic behaviour

55–95 min: Activity 6.C: Magnetic anomalies vs gravity

- Reasoning-based sketching of anomalies for idealised geometries
- Direct comparison with gravity anomalies

95–105 min: Break**105–160 min: Activity 6.D: Curie depth and magnetic crust**

- Curie temperature and magnetic crustal thickness
- Depth–wavelength relationships
- Antarctica case study
- Back-of-the-envelope heat-flow estimates

160–180 min: Synthesis and wrap-up

- What magnetism can and cannot tell us
- Connection to upcoming Fourier/spectral methods

Workshop Notes

Workshop Notes

Activity 6.A

Visualizing Magnetic Fields

Purpose

This activity develops physical intuition for magnetic fields as vector fields generated by dipoles. You will explore how field geometry depends on source orientation, strength, and proximity.

Materials

- Disc magnets
- Paper sheets
- Iron filings

Part A: Single disc magnet — vertical orientation

Place a disc magnet flat on the table and sprinkle iron filings over a sheet of paper on top.

1. Describe the symmetry of the field pattern.
2. Where are the field gradients strongest?
3. If this magnet were buried at depth, where would you expect the strongest anomaly?

Sketch:

Part B: Single disc magnet — horizontal orientation

Stand the same magnet on its edge and repeat the experiment.

1. How does the field symmetry differ from Part A?
2. Where are positive and negative lobes located?
3. How would this differ from a gravity anomaly produced by the same body?

Sketch:**Part C: Stacked disc magnets**

Stack several disc magnets and repeat Parts A and B.

1. How does stacking affect field strength?
2. Does stacking change anomaly shape or only amplitude?
3. What geological parameter does stacking mimic?

Reflection

Why are magnetic field patterns sensitive to orientation, while gravity fields are not?

Sample	Rock type	Geological context	Susceptibility (SI)	Notes
1				
2				
3				
4				
5				
6				
7				
8				

Activity 6.B

Magnetic Susceptibility of Rocks

Purpose

This activity links magnetic susceptibility to mineralogy and geological processes such as alteration, metamorphism, and weathering.

Key Reminder

Magnetic susceptibility (χ) describes how strongly a material becomes magnetised:

$$\mathbf{M} = \chi \mathbf{H}$$

Samples and Observations

Paired Comparisons

Peridotite vs Serpentinite

How does serpentinisation lead to high magnetic susceptibility?

Fresh Gabbro vs Altered Basalt

Why can alteration dramatically reduce susceptibility in mafic rocks?

Granite or Sedimentary Protolith vs Metamorphosed Equivalent

Which minerals control susceptibility in felsic or sedimentary rocks?

Synthesis

1. Why is susceptibility not a simple proxy for bulk composition?
2. Which geological processes increase susceptibility?
3. Which decrease it?

Reflection

Explain why petrology is essential for interpreting magnetic anomalies.

Activity 6.C

Magnetic Structures

Purpose

This activity mirrors previous gravity workshops. You will reason about magnetic anomaly shapes and compare them directly with gravity anomalies from the same structures.

Assume:

- Uniform susceptibility
- Induced magnetisation only
- Constant inducing field

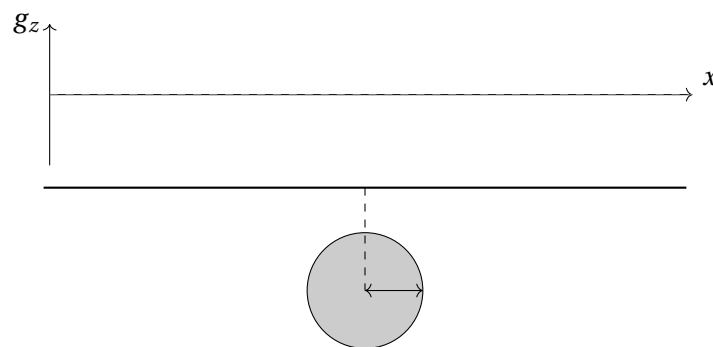
Instructions

For each structure:

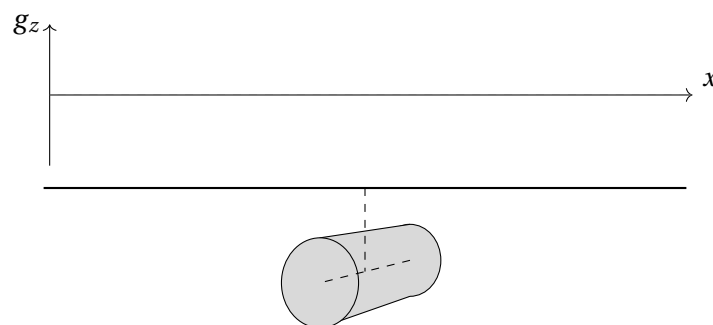
1. Sketch the expected vertical magnetic anomaly
2. Indicate sign, symmetry, width, and strongest gradients
3. State how it differs from the gravity anomaly

Structures

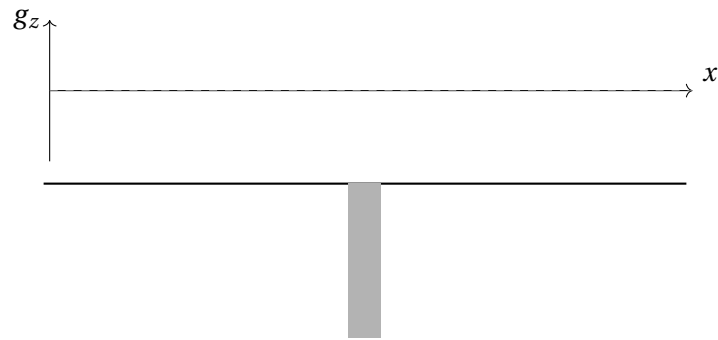
A) Buried Sphere (3D)



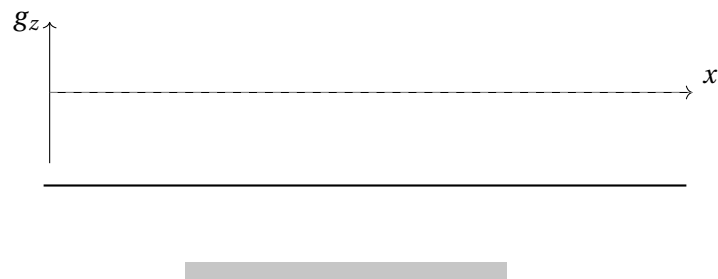
B) Buried Cylinder (2D; infinite strike)



C) Vertical Slab



D) Finite-Extent Horizontal Slab



Vertical vs Horizontal Inducing Field

Repeat one case assuming a horizontal inducing field.

1. How does the anomaly shape change?
2. Which case appears more gravity-like?
3. Why does gravity not behave this way?

Reflection

Give two reasons why magnetic interpretation is more non-unique than gravity.

Activity 6.D

Curie Depth and Magnetic Crustal Thickness

Purpose

This activity explores how temperature limits crustal magnetisation and why deep magnetic structure is long-wavelength.

Part A: Curie Depth Concept

1. What mineral dominates crustal magnetisation?
2. What happens above the Curie temperature?
3. Is Curie depth a thermal or compositional boundary?

Part B: Depth and Wavelength

Given several magnetic profiles:

1. Which correspond to shallow sources?
2. Which to deep Curie depth?
3. How does depth affect short wavelengths?

Part C: Antarctica Case Study

Consider the magnetic and crustal structure of Antarctica (Fig. 13).

Compare:

- Magnetic anomalies
- Seismic crustal thickness
- Magnetic crustal thickness

Why do seismic and magnetic thickness estimates differ?

The first-order geology of Antarctica is characterized by a stable (cratonic) region in East Antarctica and a Cenozoic rift in West Antarctica (Fig. 14).

Part D: Heat-Flow Estimate

Assume:

$$T_C = 570^\circ\text{C}, \quad k = 2.8 \text{ W m}^{-1}\text{K}^{-1}$$

$$q = k \frac{T_C}{z}$$

Estimate heat flow for a rift and a craton.

Reflection

Explain why Curie depth makes magnetics sensitive to thermal structure.

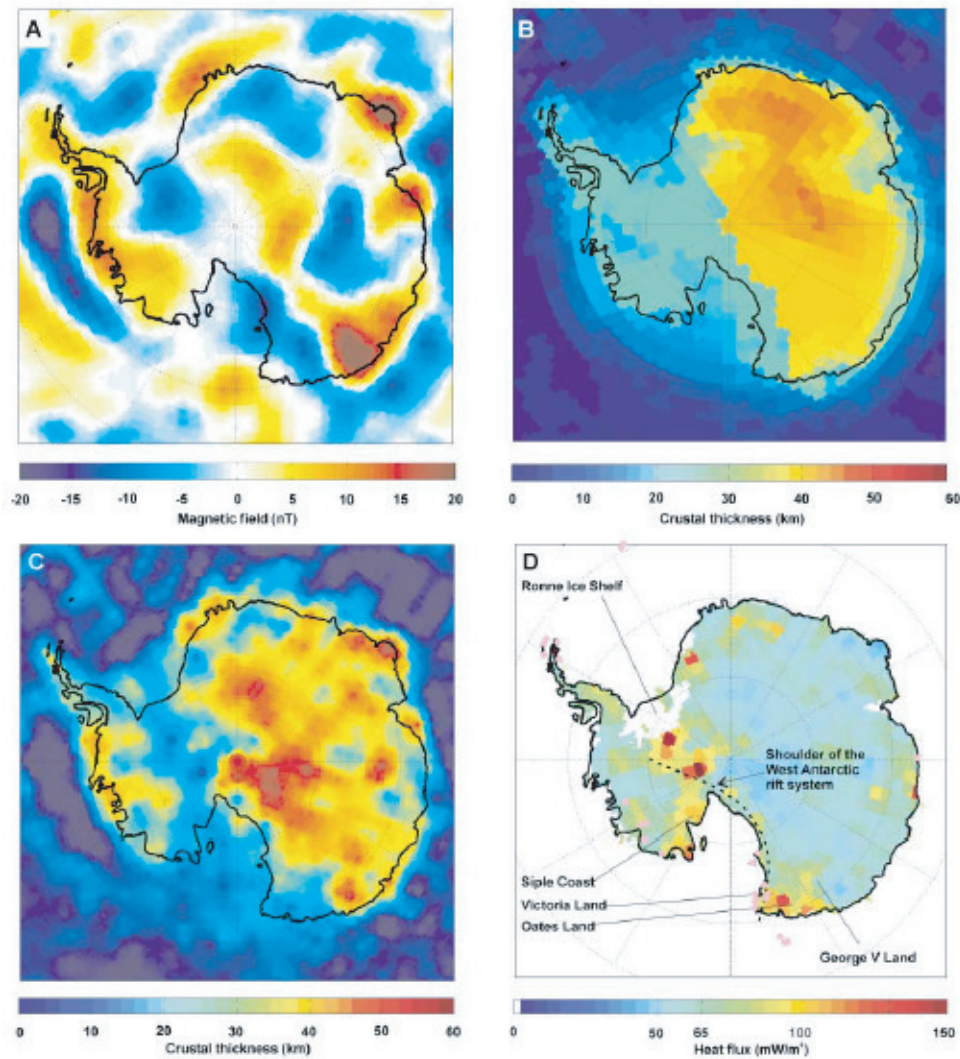
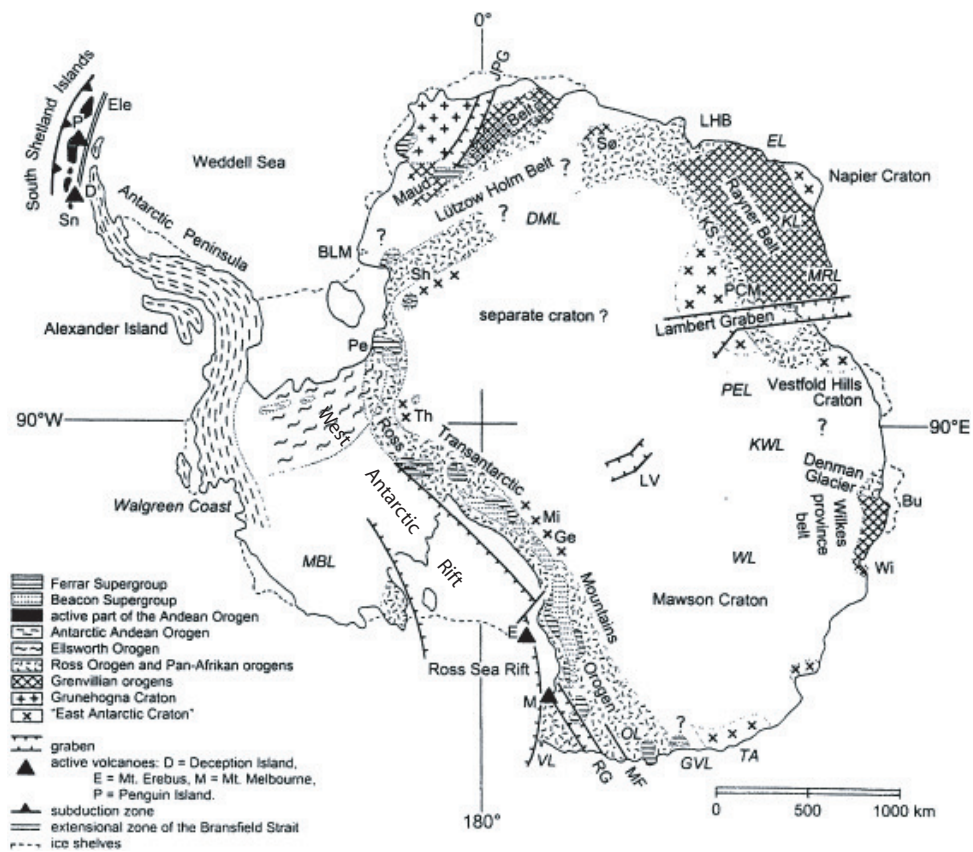


Figure 13: Estimating the Curie depth from vertical magnetic data. (a) Vertical magnetic field. (b) Crustal thickness model of Antarctica. (c) Estimate to base of magnetized layer. (d) Estimated heat flow. Figure from Maule et al. (**Science**, 2005).



Talarico and Kleinschmidt, in **Antarctic Climate Evolution**, 2008

Figure 14: Tectonic map of Antarctica from Talarico and Kleinschmidt (**Antarctic Climate Evolution**, 2008).

Week 7 — Fourier Transforms and Potential Fields

Duration: 3 hours

Setting: Workshop room (laptops for demos)

Preparation

Pre-reading: Course Notes, Ch. 17

Lecture videos:

Notes:

Purpose

This workshop develops intuition for Fourier methods in potential-field geophysics and applies them to gravity and magnetic interpretation. The emphasis is on understanding how sampling, filtering, and phase control what geological information can be recovered.

Learning Objectives

By the end of this workshop, students should be able to:

- Relate spatial wavelength to source depth and geometry
- Recognize aliasing and understand its consequences
- Interpret gravity and magnetic spectra in geological terms
- Apply upward and downward continuation appropriately
- Explain the purpose and limitations of reduction to the pole (RTP)

Session Flow (180 min)

0–15 min: Introduction and framing

- Why Fourier methods appear everywhere in potential-field geophysics
- Review of key ideas: wavelength, sampling, phase
- Overview of the day's activities

15–40 min: Demo 1: What a Fourier transform does

- Decomposing a signal into frequency, amplitude, and phase
- Phase as a measure of shift and geometry
- Padding and windowing

Key emphasis: Fourier transforms reorganize information; they do not interpret it.

40–75 min: Demo 2: Spatial sampling and aliasing

- Synthetic fold and Basin-and-Range gravity models
- Station spacing and Nyquist wavelength
- Aliasing as coherent but incorrect structure

Key emphasis: Sampling decisions set a hard limit on geological resolution.

75–95 min: Real data case study: Nevada gravity

- Complete Bouguer gravity of Nevada

- E–W and N–S power spectra
- Identification of Basin-and-Range wavelengths

95–115 min: Isostatic correction and filtering

- Airy isostasy as a physical low-pass filter
- Isostatic residual gravity
- High-pass filtering and wavelength selection

Key emphasis: Filtering encodes geological assumptions.

115–135 min: Inversion example

- Parker–Oldenburg sediment thickness inversion
- Interpretation and limitations
- Effect of aliasing on inversion results

135–165 min: Magnetic interpretation: RTP and maar geometry

- Magnetic anomalies as vector fields
- Cone/maar geometry and asymmetric anomalies
- Reduction to the pole as a phase operation

Key emphasis: RTP rotates phase; it does not filter noise.

165–180 min: Synthesis and discussion

- Gravity vs magnetics: robustness and fragility
- When Fourier methods help—and when they mislead
- Connection to upcoming assignments and field interpretation

Key Takeaways

- Fourier methods are powerful but unforgiving
- Sampling precedes interpretation
- Phase carries geometry
- Geological assumptions are embedded in every filter

Workshop Notes

Workshop Notes

Student Name: _____

Student ID: _____

Instructions (Instructor Use)

Circle the level that best reflects the student's overall performance since the previous checkpoint. Use the booklet as supporting evidence, but prioritise observed participation.

Criterion	HD	D	C	P	F
Preparation & Contribution	Insightful, advances discussion	Regular, clear reasoning	Occasional, appropriate	Basic, prompted only	Little or none
Engagement with Others	Synthesises ideas, elevates group	Builds on others' ideas	Responds to peers	Listens, limited interaction	Disengaged
Curiosity & Thinking	Explores ambiguity, asks deep questions	Engages with complexity	Asks genuine questions	Attempts answers when prompted	Avoids engagement
Booklet Evidence	Complete, thoughtful, annotated	Mostly complete, good detail	Adequate, some gaps	Minimal effort evident	Sparse or missing

Overall Performance

HD

D

C

P

F

Notes (optional)

Week 8 — Geothermics: Steady-State Heat Flow

Duration: 3 hours

Setting: Workshop room

Preparation

Pre-reading: Course Notes, Ch. 10

Lecture videos:

Notes:

Purpose

This workshop uses conservation-of-energy reasoning to build intuition for steady-state geotherms: how thermal conductivity contrasts refract isotherms and redistribute gradients, and how crustal heat production controls geotherm curvature and deep temperatures.

Learning Outcomes

By the end of this workshop, students should be able to:

- Use conservation of energy to reason about steady-state geotherms.
- Explain how thermal conductivity contrasts redistribute isotherms.
- Interpret geotherm curvature in terms of crustal heat production.
- Relate conductivity and heat production variations to geothermal potential.

Session Flow (180 min)

0–15 min: Conceptual Framing

- Steady-state heat flow as an energy bookkeeping problem.
- Review Fourier's law and the meaning of heat flow.
- Emphasize the separation of roles:
 - conductivity controls *gradients*,
 - heat production controls *curvature*.

15–45 min: Activity 8.A: Refraction Exercise I: Insulating Sediments

Setup Compare two crustal columns with identical basal heat flow:

- Case A: crystalline crust exposed at the surface.
- Case B: crystalline crust overlain by a low-conductivity sedimentary layer.

Activities

- Predict temperature gradients in sediments vs basement.
- Sketch qualitative isotherms for each case.
- Identify where isotherms compress and where they stretch.

Key Concepts

- Heat flow conserved across layers.
- Lower conductivity → steeper gradient.
- Sediments act as a thermal blanket.

Geological Discussion

- Implications for geothermal gradients measured in boreholes.
- Relative geothermal potential of sedimentary basins vs exposed cratons.

45–80 min: Activity 8.B: Refraction Exercise II: Layered Crustal Materials

Focus Thermal refraction arising from conductivity contrasts within the crust.

Examples

- Shales (low k) vs quartzites (high k).
- Cyclic sedimentary layering preserved through metamorphism.

Activity

- Apply $q = -k\nabla T$ across layered media.
- Examine how gradients vary layer by layer.
- Discuss effective anisotropy in cyclically layered strata.

Connection

- Tie back to Week 2 physical-properties mixing demos.
- Discuss refraction in tilted and folded strata.

80–90 min: Break

90–125 min: Activity 8.C: Energy Conservation and Geotherm Curvature

Central Question How can two terranes with identical surface heat flow have different Moho temperatures?

Steps

- Write the energy balance:

$$q_0 = q_b + \int_0^{z_c} A(z) dz$$

- Infer relative crustal heat production given different basal inputs.
- Introduce the steady-state solution and the curvature term:

$$T(z) = T_0 + \frac{q_0}{k} z - \frac{A}{2k} z^2$$

Key Insight Curvature differences imply different deep temperatures even when surface gradients match.

125–160 min: Activity 8.D: Why Is Archean Heat Production So Low?

Observation Decay alone predicts high early heat production, yet preserved Archean crust shows limits.

Guided Discussion Topics

- Differentiation and partitioning of LILEs (K, Th, U).
- Selective erosion of felsic upper crust.
- Reworking: remelting and redistribution of older crust.
- Foundering of dense mafic roots.
- Secular mantle thermal evolution.

160–180 min: Synthesis

- Summarize roles of k and A .
- Connect observations to upcoming steady-state problem set.
- Preview transition to transient heat flow.

Workshop Notes

Workshop Notes

Activity 8.A

Thermal Refraction by Insulating Sediments

Purpose

To build intuition for how low-conductivity surface layers redistribute temperature gradients without changing the total heat flow.

Geological Context

Consider two continental regions with identical basal heat flow from the mantle:

- Region A: crystalline crust exposed at the surface.
- Region B: crystalline crust overlain by a thick sedimentary basin.

The sediments are dominated by shales and porous clastic rocks with lower thermal conductivity than basement rock.

Assumptions

- Steady-state heat flow.
- No internal heat production in sediments.
- Heat flow is vertical and laterally uniform.

Tasks

1. Using Fourier's law,

$$q = -k \frac{dT}{dz},$$

explain qualitatively how the temperature gradient differs between sediments and basement.

2. Sketch temperature versus depth for both regions on the same axes. Clearly label:
 - sediment thickness,
 - basement,
 - relative steepness of gradients.
3. Indicate where isotherms are compressed and where they are stretched.
4. Which region would exhibit a higher temperature at the sediment–basement interface? Explain why.

Discussion

- How does the presence of sediments affect measured geothermal gradients?
- Which region would you expect to have higher geothermal potential?
- Why might sedimentary basins be favorable geothermal targets?

Activity 8.B

Thermal Refraction in Layered Crustal Rocks

Purpose

To explore thermal refraction caused by conductivity contrasts within the crust, particularly in sedimentary and metamorphic layering.

Geological Context

Many crustal sections contain cyclic layering inherited from sedimentary environments, such as alternating shales and quartz-rich sandstones that later become quartzites.

- Shales: low thermal conductivity.
- Quartzites: high thermal conductivity.

This layering can persist through burial, deformation, and metamorphism.

Tasks

1. Consider a vertical sequence of alternating shale and quartzite layers. Assume constant heat flow through the sequence.
2. Using

$$q = -k\nabla T,$$

explain how the temperature gradient differs in shale layers compared to quartzite layers.

3. Sketch a qualitative temperature–depth profile showing changes in slope across layers.
4. Now consider the same layered sequence tilted relative to vertical heat flow. Describe how thermal refraction will distort isotherms.

Extension

- How can cyclic layering create an effectively anisotropic thermal conductivity?
- How might folding or tilting further modify the thermal field?

Connection

Relate this behavior to the physical-properties mixing demonstrations discussed earlier in the course.

Activity 8.C

Energy Conservation and Geotherm Curvature

Purpose

To understand how crustal heat production alters lithospheric temperatures even when surface heat flow is unchanged.

Scenario

Two continental terranes have identical surface heat flow, q_0 , but different amounts of heat supplied from the mantle at the base of the crust.

Tasks

1. Write the steady-state energy balance for the crust:

$$q_0 = q_b + \int_0^{z_c} A(z) dz$$

2. If Terrane A has a larger basal heat flow (q_b) than Terrane B, which must have higher integrated crustal heat production? Explain.
3. Consider the steady-state geotherm:

$$T(z) = T_0 + \frac{q_0}{k} z - \frac{A}{2k} z^2$$

Identify which term controls curvature.

4. Without performing calculations, explain how greater curvature affects:
 - temperature at the Moho,
 - depth of intersection with the mantle adiabat.

Key Insight

Explain why surface gradients alone are insufficient to determine deep crustal temperatures.

Activity 8.D

Archean Heat Production

Purpose

To connect radioactive decay, crustal differentiation, and geological processes to the observed heat production of ancient continental crust, we'll investigate the following question "Why is Archean heat production so low?"

Radiogenic decay

The concentration of a radioactive isotope is given by

$$C(t) = C_0 e^{-\lambda t},$$

and the corresponding heat production is

$$A(t) = \sum_i H_i \lambda_i C_i(t),$$

where H_i is the heat released per decay and λ_i is the decay constant.

Observation

Simple decay arguments predict higher heat production in the Archean than today, yet preserved Archean crust shows relatively modest values.

Tasks

For each mechanism below, explain qualitatively how it could reduce the heat production of preserved crustal rocks:

1. Differentiation and melt extraction of LILEs (K, Th, U).
2. Selective erosion of felsic upper crust.
3. Reworking: remelting and redistribution of existing crust.
4. Foundering of dense mafic lower crust or arc roots.
5. Secular evolution of mantle temperature.

Differentiation

Concept: Heat-producing elements (K, Th, U) behave as incompatible elements and strongly partition into melts.

Let $D \ll 1$ be the bulk partition coefficient of a heat-producing element. For fractional melting,

$$C_{\text{solid}} = C_0 (1 - F)^{(1/D-1)},$$

where F is the melt fraction. You could think of this simply as:

$$C_{\text{crust}} \approx \frac{C_{\text{mantle}}}{D} F$$

Selective erosion

Concept: Heat production is concentrated in felsic upper crust, which is preferentially eroded.

Treat the crust as two reservoirs. Let A_u be upper crust heat production. If erosion removes heat-producing material at rate E ,

$$\frac{dA_u}{dt} = -EA_u.$$

Solution:

$$A_u(t) = A_{u0}e^{-Et}$$

Reworking

Reworking here will be defined as reheating—past the closure temperatures of geochronometers—and remelting of existing crust rather than net addition of new crust.

Concept: Old crust is repeatedly melted, mixed, and redistributed during later tectonic events.

Schematic formulation: Model reworking as a mixing process. Let a fraction R of crust be reprocessed per event.

$$A_{n+1} = (1 - R)A_n + RA_{\text{new}},$$

where A_{new} reflects heat production typical of newly generated crust.

Foundering

Concept: Dense, mafic, heat-poor material is preferentially removed from the lithosphere.

Let the lower crustal mass is reduced over time. If mafic lower crust is removed at rate F_d ,

$$\frac{dM_{lc}}{dt} = -F_d M_{lc}.$$

The average crustal heat production becomes:

$$\bar{A}(t) = \frac{A_u M_u + A_{lc} M_{lc}(t)}{M_u + M_{lc}(t)}.$$

Secular cooling

Concept: The mantle was hotter early, affecting melt production rather than radiogenic inventory.

Mantle potential temperature declines approximately as:

$$T_m(t) = T_{m0}e^{-t/\tau},$$

where $\tau \sim 1\text{--}2$ Ga is a thermal timescale. Melt production gives:

$$F(t) \propto T_m(t) - T_{\text{solidus}}.$$

Synthesis Questions

1. Which processes change the *distribution* of heat-producing elements?
2. Which processes affect what crust is *preserved*?
3. Why does radioactive decay alone fail to predict observed Archean heat production?

Which processes primarily:

Process	Primary Physical Mechanism	Effect on Heat-Producing Elements (K, Th, U)	Primary Geological Consequence
Radiogenic decay	Exponential decay of radioactive isotopes	Monotonic decrease of heat production with time	Higher heat production in the past; explains first-order time trend only
Differentiation and melt extraction	Incompatible elements partition strongly into melts ($D \ll 1$)	Rapid transfer of LILEs from mantle to crust early in Earth history	Early enrichment of crust; depletion of mantle
Selective erosion	Preferential removal of felsic, heat-rich upper crust	Progressive loss of the most heat-producing crustal components	Older crust appears cooler than expected from decay alone
Reworking (remelting and redistribution)	Repeated partial melting and crustal mixing	Dilution and homogenization of extreme heat production signatures	Preserved Archean crust is rarely pristine
Foundering of mafic lower crust	Removal of dense, heat-poor lower lithosphere	Bias toward preservation of felsic, buoyant crust	Thermally stable, thick continental lithosphere
Secular mantle thermal evolution	Early high mantle temperatures enhance melt production	Controls efficiency of LILE extraction rather than inventory	Strong early differentiation followed by declining crustal growth

- redistribute heat-producing elements?
- remove heat-producing material from the crust?
- bias what crust survives to the present day?

Reflection

Why is radioactive decay alone insufficient to explain Archean heat production patterns?

Concept Takeaway: Heat Production, Curvature, and Preservation

Radiogenic decay alone predicts that heat production should decrease smoothly with time, but observed Archean crust shows much lower heat production than this simple expectation.

The heat production of preserved continental crust reflects the combined effects of:

- *redistribution* of heat-producing elements through differentiation and melting,
- *removal* of heat-rich material by erosion and tectonic processes, and
- *preservation bias*, where buoyant, felsic, and thermally stable crust is more likely to survive.

In steady-state geotherms, these processes matter because crustal heat production controls the *curvature* of the geotherm:

$$T(z) = T_0 + \frac{q_0}{k}z - \frac{A}{2k}z^2.$$

Two regions with identical surface heat flow can therefore have very different Moho temperatures, lithospheric thicknesses, and long-term stability depending on how heat-producing elements were redistributed and preserved through time.

Week 9 — Geothermics: Transient Heat Flow

Duration: 3 hours

Setting: Workshop room

Preparation

Pre-reading: Course Notes, Ch. 11

Lecture videos:

Notes:

Purpose

This workshop develops intuition for transient heat flow: how thermal diffusivity sets the timescales and length scales over which thermal signals propagate, how half-space and plate cooling models describe the thermal evolution of oceanic lithosphere, and how latent heat modifies cooling in Stefan problems.

Learning Outcomes

Students will:

- Understand how thermal diffusivity controls timescales.
- Compare thermal and chemical diffusion processes.
- Interpret half-space and plate cooling models.
- Use Fourier series to conceptualize thermal evolution.

Session Flow (180 min)

0–45 min: Activity 9.A: Thermal Diffusivity, Thermal Length, and Timescales

- Review $\kappa = k/(\rho c_p)$.
- Emphasize modest variation in crustal rocks.
- Contrast sediments, rock, and ice.
- Penetration depth scaling: $z \sim \sqrt{\kappa t}$.
- Borehole paleothermometry vs ice-core records.
- Depth–resolution tradeoffs.

45–75 min: Activity 9.B: Thermal vs Chemical Diffusion

- Both obey $\sqrt{Dt}$ scaling.
- Chemical diffusion often multi-component.
- Zoning persistence vs thermal smoothing.

75–85 min: Break

85–120 min: Activity 9.C: Half-Space Cooling

- Assumptions and boundary conditions.
- Growth of thermal boundary layer.
- Oceanic lithosphere as a thermal clock.

120–150 min: Activity 9.D: Plate Cooling and Fourier Series

- Fixed plate thickness as boundary condition.

- Fourier series as superposition of thermal modes.
- Mode decay times and geological interpretation.

150–180 min: Activity 9.E: Stefan Problems in Geology

- Latent heat as effective heat capacity.
- Cooling times of dikes vs sills.
- When analytic solutions break down.

Workshop Notes

Workshop Notes

Material	Thermal Conductivity k (W m ⁻¹ K ⁻¹)	Thermal Diffusivity κ (m ² s ⁻¹)	Thermal Diffusivity κ (km ² My ⁻¹)
Ice (glacial / ice sheet average)	1.6–2.0	$(0.7\text{--}0.9) \times 10^{-6}$	22–28
Clay-rich sediments	0.6–1.5	$(0.2\text{--}0.5) \times 10^{-6}$	6–16
Granite	2.5–3.5	$(0.8\text{--}1.2) \times 10^{-6}$	25–38
Basalt / Gabbro	1.7–2.2	$(0.6\text{--}1.0) \times 10^{-6}$	19–32
Peridotite (upper mantle)	3.3–4.5	$(1.0\text{--}1.5) \times 10^{-6}$	32–47

Activity 9.A

Thermal Diffusivity, Thermal Length, and Timescales

Purpose

To understand transient heat flow in terms of:

- how far thermal signals propagate in a given time,
- how long it takes thermal changes to affect a given depth,
- and how the *thermal length scale* controls signal shape and timing.

Background

Thermal diffusivity is defined as

$$\kappa = \frac{k}{\rho c_p},$$

where k is thermal conductivity, ρ is density, and c_p is specific heat.

Values are representative ranges. Thermal conductivity and diffusivity vary with temperature, porosity, mineralogy, fluid content, and fabric. Diffusivities are calculated using $\kappa = k/(\rho c_p)$ and rounded for scaling arguments rather than precise modeling.

For diffusion-dominated heat transfer, the characteristic depth over which temperature changes propagate scales as

$$z \sim 2\sqrt{\kappa t}.$$

This thermal length sets both:

- the depth to which temperature changes have propagated after time t ,
- the timescale required for a thermal perturbation at depth to influence shallower levels.

Error Function Solution

For instantaneous heating or cooling at a boundary, the temperature evolution is given by the error function solution:

$$T(z, t) = T_s + (T_b - T_s) \operatorname{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

Important: The argument of the error function,

$$\eta = \frac{z}{2\sqrt{\kappa t}},$$

controls the *shape and timing* of the thermal signal.

Tasks

1. For timescales of 10^3 , 10^5 , and 10^7 years, describe how the depth of thermal influence would differ between rock and ice.
2. Explain why borehole paleothermometry in rock requires much greater depths to recover long-term climate signals than ice-core records.

Tasks

A. Distance Given Time

1. For a fixed time t , describe how temperature varies with depth z as the value of η increases.
2. Explain why temperature changes are largest for $\eta \ll 1$ and become negligible for $\eta \gg 1$.
3. Using the thermal length ℓ_T , estimate the depth to which a surface temperature change has propagated after a given time.

B. Time Given Distance

1. Rearranging the thermal length relation,

$$t \sim \frac{z^2}{\kappa},$$

explain what sets the timescale for heat to travel a distance z .

2. Consider a plume impinging on the base of the lithosphere. If the lithosphere is x km thick, what controls how long it takes for this thermal perturbation to influence surface temperatures?
3. Following delamination of a dense crustal root, heat is supplied from below. Explain why crustal heating is delayed rather than instantaneous.

C. Interpreting “Arrival” of Heat

1. Does heat arrive at a specific time, or does it diffuse gradually? Use the error function shape to justify your answer.
2. For $\eta \approx 1$, what fraction of the total temperature change has occurred? What does this imply about defining a thermal response time?
3. Discuss why the concept of a sharp thermal front is inappropriate for diffusion-dominated heat transport.

Geological Interpretation

- Thick lithosphere acts as a strong thermal filter, delaying surface response to basal heating.
- Thermal signals generated at depth are attenuated and spread in time as they propagate upward.
- The thermal length scale controls which geological processes can be observed at the surface and on what timescales.

Key Insight

In diffusion problems, time and distance are interchangeable through the thermal length $\ell_T = 2\sqrt{\kappa t}$. Heat does not “arrive” at a depth at a single moment; instead, the error function describes a gradual, scale-dependent thermal response controlled by diffusivity, distance, and boundary conditions.

Activity 9.B

Thermal vs Chemical Diffusion

Purpose

To compare transient thermal diffusion with chemical diffusion processes familiar from igneous and metamorphic petrology.

Diffusion Framework

Both thermal and chemical diffusion can be written in the general form

$$\frac{\partial \phi}{\partial t} = D \nabla^2 \phi,$$

where ϕ represents temperature or chemical concentration, and D is a diffusivity.

Tasks

1. Explain why both thermal and chemical diffusion act to smooth spatial gradients over time.
2. Chemical diffusion in minerals often preserves sharp zoning. Give two reasons why sharp compositional gradients may survive despite diffusion.
3. Describe how multi-component diffusion differs from thermal diffusion involving a single scalar field.

Comparison

- Thermal diffusion: single field, relatively uniform diffusivity.
- Chemical diffusion: multi-component, potentially rate-limited by the slowest species.

Reflection

Why is heat generally a poorer long-term recorder of geological processes than composition?

Activity 9.C

Half-Space Cooling of the Oceanic Lithosphere

Purpose

To develop physical intuition for transient cooling of the oceanic lithosphere using the half-space cooling model.

Model Assumptions

- The mantle is initially at uniform temperature.
- Surface temperature is suddenly reduced and held fixed.
- No fixed lithospheric thickness is imposed.

Solution Form

The temperature evolution is given by

$$T(z, t) = T_m \operatorname{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

Tasks

1. Sketch temperature–depth profiles for increasing times.
2. Identify the thermal boundary layer and describe how its thickness evolves.
3. Estimate how the depth to a given isotherm scales with seafloor age.

Geological Interpretation

- Oceanic lithosphere thickens thermally with age.
- Heat flow decreases as the plate cools away from mid-ocean ridges.

Key Insight

Explain why the half-space cooling model predicts unbounded lithospheric thickening.

Activity 9.D

Plate Cooling and Fourier Series

Purpose

To understand how imposing a fixed lithospheric thickness modifies transient cooling and why Fourier series naturally arise in plate cooling models.

Concept

Plate models introduce a basal boundary condition at a fixed depth, representing the lithosphere–asthenosphere boundary.

Mathematical Form

The temperature field can be expressed as a Fourier series:

$$T(z, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi z}{L}\right) \exp\left(-\frac{n^2 \pi^2 \kappa t}{L^2}\right).$$

Tasks

1. Explain how the index n relates to thermal wavelength.
2. Why do short-wavelength modes decay faster than long-wavelength modes?
3. Compare thermal evolution predicted by plate cooling and half-space cooling.

Geological Interpretation

- Plate thickness limits lithospheric growth.
- Explains flattening of bathymetry and heat flow for old oceanic plates.

Connection

Relate Fourier modes to spectral filtering concepts used previously in gravity and magnetic field analysis.

Activity 9.E

Stefan Problems and Cooling of Magma Bodies

Purpose

To apply transient heat flow concepts to the crystallization and cooling of magma bodies within the crust.

Physical Problem

A magma body emplaced into cooler host rock cools conductively while releasing latent heat at the solidification front.

Key Feature

The solid–liquid boundary migrates with time, making this a Stefan problem.

Tasks

1. Explain how latent heat affects the cooling rate of a magma body.
2. Compare expected cooling timescales for a narrow dike and a thick sill.
3. Identify geological situations in which analytic Stefan solutions are likely to fail.

Geological Extensions

- Magma convection.
- Episodic recharge of magma.
- Crystal-rich mush zones.

Key Insight

Geometry and diffusivity often dominate cooling timescales more strongly than radiogenic heat production.

Week 10 — Geothermics: Finite Differences

Duration: 3 hours

Setting: Workshop room

Preparation

Pre-reading: Course Notes, Ch. 18

Lecture videos:

Notes:

Purpose

This workshop builds intuition for finite difference methods through physical reasoning and interactive activities. Students first derive the heat equation from energy conservation, then experience how numerical models work by acting as nodes that update temperature based on their neighbours. The workshop then explores how boundary conditions represent geological assumptions and how numerical stability reflects limits on how heat can physically propagate. By the end, students should understand that finite difference methods are simply local energy balance, that heat flow emerges from neighbour interactions, and that both boundary conditions and stability have direct physical meaning.

Learning Outcomes

Students will:

- Understand finite difference equations as local statements of energy conservation.
- Develop intuition for how numerical solutions evolve through neighbour interactions.
- Interpret boundary conditions as geological assumptions.
- Understand numerical stability as a limit on physical information transfer.

Session Flow (180 min)

0–30 min: Activity 10.A: From Conservation of Energy to Finite Differences

Concept

Finite difference equations arise from applying conservation of energy to small rock volumes.

- Nodes represent local rock volumes.
- Heat flux enters and leaves each node.
- Heat production adds energy locally.

The equation is *not a numerical trick*. The Laplacian as a measure of curvature (difference from neighbours). This is just bookkeeping of energy.

30–65 min: Activity 10.B: Finite Differences as a Physical Process: Human Finite Difference Solver

Concept

Numerical models evolve through local interactions between neighbouring nodes.

Discuss:

- smoothing of temperature
- spreading of heat

- role of gradients

65–85 min: Activity 10.C: Boundary Conditions as Geological Assumptions

Concept: Boundary conditions determine what physical system the model represents.

- Fixed temperature (Dirichlet)
- Fixed heat flux (Neumann)

85–95 min: Break

95–120 min: When Analytic Solutions Break Down

Concept

- Analytic solutions require simplifications:
 - constant properties
 - simple geometries
 - simple boundary conditions
- Real geology violates these assumptions.

Discussion prompts

- What makes geological systems difficult to solve analytically?
- Why do irregular boundaries force numerical approaches?

Your Iberia exercise on the problem has several issues that make it impossible to develop an analytic form, including variable sediment thickness, lateral structure, varying boundary condition.

120–160 min: Activity 10.D: Numerical Stability and Information Transfer

Concept Stability limits impose a “speed limit” on thermal information:

$$\Delta t < \frac{\Delta x^2}{2\kappa}$$

- Heat propagates gradually through diffusion.
- Numerical schemes must respect this.
- small vs large time steps
- what happens when updates are too large
- stability = causality
- information cannot skip nodes

Instability is not just numerical—it is physically impossible behaviour.

160–180 min: Wrap-Up and Synthesis

Key connections

- Finite differences = local energy balance + neighbour interaction
- Boundary conditions = geological assumptions
- Stability = physical constraint on information transfer

Discussion

- How does the numerical method compare to analytic solutions?
- What advantages do numerical models provide?

- Where might these methods be applied beyond heat flow?

Preview

- Extension to 2-D systems
- Application to real datasets (Iberia Abyssal Plain)

Workshop Notes

Workshop Notes

Activity 10.A

From Conservation of Energy to Finite Differences

Purpose

To understand finite difference heat flow equations as direct consequences of energy conservation applied to small volumes of rock.

Consider a vertical column of sediment divided into small layers. Heat flows between layers, and each layer can gain or lose energy depending on its neighbours and any internal heat production.

Rather than solving a continuous equation, we will track how heat moves between *discrete points*.

Part A: Physical intuition

- If a layer is hotter than the ones above and below it, does it gain or lose heat?
- If heat is being produced inside the layer, what happens over time?

Part B: Setting up the system

- Sketch a one-dimensional grid of nodes labeled T_{i-1} , T_i , and T_{i+1} .
- Indicate the spacing Δz .

Part C: Heat flux

Heat flows according to Fourier's law:

$$q_z = -k \frac{\partial T}{\partial z} \quad (9)$$

Tasks:

- Write an expression for heat flowing *into* node i .
- Write an expression for heat flowing *out of* node i .

Part D: Energy conservation

Energy conservation for a small volume gives:

$$\text{Rate of change} = \text{in} - \text{out} + \text{produced}$$

Task: Combine your expressions to derive:

$$\frac{\partial T_i}{\partial t} = \kappa \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta z^2} + \frac{A_i}{\rho c_p} \quad (10)$$

Part E: Interpretation

Explain in words:

- What does the term $T_{i+1} - 2T_i + T_{i-1}$ represent?

- What role does heat production A_i play?
- What happens if $A_i = 0$?

Key Insight

Finite difference equations are not numerical tricks—they are simply energy conservation applied locally.

Activity 10.B

Solving the Heat Equation as a Human Computer

Purpose

To develop intuition for finite difference methods by acting as the nodes in a numerical heat flow model.

In a numerical model, each point (node) updates its temperature using only information from its neighbours.

In this activity, you will act as the nodes in a finite difference model.

Box .2: Human Computer

Before electronic computers existed, a *computer* was literally a person employed to carry out calculations by hand or with mechanical calculators. Large groups of human computers were used in astronomy, navigation, ballistics, engineering, and later for the numerical solution of differential equations. By the late 19th and early 20th centuries, many of these computing groups were composed largely of women. Examples include the astronomical computers working for the *Nautical Almanac* and later teams at observatories, government laboratories, and research institutions.

During World War II, the scale of numerical work *exploded* (pun intended). Hundreds of human computers computed ballistic firing tables and solved systems of finite-difference equations for artillery trajectories, fluid flow, and other military problems. Post-WWII, human computers were also instrumental to the US space program. If you've seen the movie *Hidden Figures*, you learned about some of these remarkable women who helped make going to the Moon possible by producing many of the calculations needed to determine flight plans, rocket burn parameters, launch windows, and safe re-entry trajectories.

Finite-difference methods were especially important because many physical problems could not be solved analytically. Human computers discretized differential equations onto grids and repeatedly computed numerical updates row by row—an extremely laborious process that helped motivate the development of electronic computers. The first six programmers of the world's first general-purpose electronic computer, ENIAC, were women who had previously worked as human computers.

David Alan Grier, **2005**, *When Computers Were Human*, Princeton University Press.

Kathy Kleiman, **2022**, *Proving Ground: The Untold Story of the Six Women Who Programmed the World's First Modern Computer*.

David Alan Grier, **2001**, *Human computers: the first pioneers of the information age*, **Endeavour** 25(1).

Setup

- Each student is assigned a position in a line.
- Each student is given an initial temperature value T_i .
- The two students at the ends of the line represent **fixed boundary conditions**.

Update rule

At each step, interior nodes update their temperature according to:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} \quad (11)$$

This means:

- you look at your two neighbours,
- take the average,
- and update your value.

Part A: Prediction

Before starting, discuss:

- What happens to a sharp temperature spike over time?
- Will the system become smoother or less smooth?

Part B: Iteration

Repeat the following for several steps:

1. Show your value to your neighbours.
2. Compute your new value.
3. Update simultaneously.

Record the evolution of the system.

Part C: Observations

Discuss:

- Where do the largest changes occur?
- What happens to temperature peaks?
- How does heat move through the system?

Part D: Interpretation

Explain in words:

- Why does the system become smoother over time?
- What role do neighbouring nodes play?
- How does this relate to diffusion?

Key Insight

Heat flow is a local process: each point evolves based on its neighbours.

Optional Extension 1: Heat Production

Introduce a constant heat source at one node.

New update rule:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} + S \quad (12)$$

where S is a constant heat input.

Tasks:

- What happens to the system over time?
 - Does it still become uniform?
 - What steady-state behaviour emerges?
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Optional Extension 2: Two-Dimensional Grid

Arrange students in a grid.

Update rule:

$$T_{i,j}^{\text{new}} = \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}}{4} \quad (13)$$

Tasks:

- Observe how a hot region spreads in two dimensions.
- Compare to the one-dimensional case.
- Describe the shape of the temperature field over time.

Final Reflection

- In what sense are you “solving the heat equation”?
- Why does this method require many iterations?

Activity 10.C

Boundary Conditions and Geological Meaning

Purpose

To understand how boundary conditions encode geological assumptions in numerical models.

A numerical model does not “know” geology—it only knows equations and boundary conditions.

Boundary conditions determine what kind of Earth your model represents.

Part A: Grounding in real systems

Consider the following settings:

- Ocean floor beneath deep water
- Continental lithosphere
- A hot magma body cooling in the crust

Task: For each case, describe what controls the temperature at the boundaries.

Part B: Types of boundary conditions

Common thermal boundary conditions:

- Fixed temperature (Dirichlet)
- Fixed heat flux (Neumann)

Task: Match each type of boundary condition to a geological situation.

Part C: Comparing assumptions

Explain how the model changes if you prescribe:

- surface temperature
- surface heat flow

What is fixed in each case? What is allowed to vary?

Part D: Geological consequences

Compare appropriate boundary conditions for:

- oceanic lithosphere
- continental lithosphere

Part E: Critical thinking

Task: Explain why inappropriate boundary conditions can produce physically misleading results—even if the numerical solution is mathematically stable.

Reflection

If two models use the same equation but different boundary conditions, can they produce completely different results? Why?

Key Insight

Boundary conditions are where geological assumptions enter a numerical model.

Activity 10.D

Stability, Time Stepping, and Information Transfer

Purpose

To develop physical intuition for numerical stability in explicit finite difference schemes.

In a time-stepping model, temperature is updated step-by-step. Each step uses information from neighbouring nodes.

Think of heat as spreading gradually through the system.

Part A: Prediction

Consider two cases:

- very small time steps
- very large time steps

What do you expect to happen in each case?

Part B: Physical reasoning

Task: Explain why heat cannot move arbitrarily far in a single time step.

What determines how quickly heat diffuses through a material?

Part C: Stability condition

For a one-dimensional explicit scheme:

$$\Delta t < \frac{\Delta z^2}{2\kappa} \quad (14)$$

Tasks:

- Why does the allowable time step depend on Δz ?
- Why does it depend on thermal diffusivity κ ?

Part D: Interpretation

Task: Interpret the stability condition as a limit on how fast thermal information can move between nodes.

Why can heat not “jump” over nodes in a single time step?

Part E: Geological interpretation

Explain:

- Why are sedimentary basins (low κ) easier to model?
- Why do high-diffusivity materials require smaller time steps?

Key Insight

Numerical instability occurs when the model allows heat to propagate faster than physically possible.